QUANT SERIES · PART 4 OF 5

Stochastic Calculus, Step by Step

Part 4: continuous-time pricing and numerical methods



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 4 OF 5

Stochastic Calculus, Step by Step

Part 4: Continuous-time pricing and numerical methods, with Python

Rates with Bip

Stochastic Calculus, Step by Step. Part 4: Continuous-time pricing and numerical methods, with Python.

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One set of numbers runs through the whole course: a share priced at $S_0 = 100$, a strike $K = 105$, a risk-free rate $r = 3\%$ (continuously compounded), a volatility $\sigma = 20\%$ and a horizon $T = 1$ year; where a lesson needs the share's real-world expected return, it is $\mu = 8\%$. They are made up for teaching, and every random path is simulated with a fixed seed. There is no market data.

The exercises are at the end of each lesson. Their full solutions, and the answers to the self-test questions, are in the companion volume, Stochastic Calculus, Step by Step, Part 4: Solutions.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only), one per lesson, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This is Part 4 of *Stochastic Calculus, Step by Step*, a course in five parts that goes from school calculus to the Black-Scholes formula, Feynman-Kac and the numerical methods banks use to price options. Part 1 laid the mathematical foundations, Part 2 built option pricing on a coin-toss tree, and Part 3 built the calculus of Brownian motion. **Part 4 is where it pays off**: the Black-Scholes equation and formula, derived in full; every Greek; hedging, simulated until you believe it; Girsanov's theorem and risk-neutral pricing, proved; Feynman-Kac; and the methods that take over when there's no formula.

LESSON	TOPIC	WHAT YOU TAKE AWAY
19	The Black-Scholes equation	The hedge $\Delta = V_S$, and why μ vanishes
20	The Black-Scholes formula	Every line of the derivation
21	The Greeks	Delta, gamma, theta, vega, and the balance between them
22	Checking the formula numerically	Monte Carlo error bars, tree convergence, the American put
23	The trinomial tree	Moment matching, and what the third branch buys
24	Delta hedging in practice	The error's size and cause; hedging with the wrong volatility
25	Girsanov's theorem	Changing the drift by reweighting the paths
26	Risk-neutral pricing	The pricing formula, the hedge, the two fundamental theorems
27	Change of numéraire	What $N(d_+)$ and $N(d_-)$ really are; Margrabe's formula
28	The Feynman-Kac theorem	The bridge between PDEs and expectations
29	Finite differences	Explicit and implicit schemes, and the stability limit
30	Choosing a method	Which tool, when, and why

Who this book is for

As in the earlier parts, this book is written for readers learning on their own, without a teacher to fill the gaps. Every new idea is **defined** before it is used, **explained** in words, tried on a small example, and then **derived** one line at a time, with the reason for each

line. When a step uses an earlier lesson, the lesson is named.

You need Part 3 above all: Itô's lemma (Lesson 16), geometric Brownian motion (Lesson 17) and the exponential martingale and the martingale test (Lesson 18). From Part 2 you need replication and the risk-neutral idea (Lessons 9 and 11). From Part 1, the normal distribution and completing the square (Lesson 4) and partial derivatives (Lesson 2).

One set of numbers

Part 4 uses the course's numbers: a share at $S_0 = 100$, a strike $K = 105$, a volatility $\sigma = 20\%$, a risk-free rate $r = 3\%$ (continuously compounded) and a horizon of $T = 1$ year. New in this part is a real-world expected return $\mu = 8\%$, different from r, because the point of several lessons is to watch it disappear. The market price of risk is then $\theta = (\mu - r)/\sigma = 0.25$. The call you will meet on nearly every page is worth 7.1281, the put 9.0248. Every random path is simulated with a fixed seed.

How each lesson is built

Every lesson follows the same steps as before: **why this lesson**, **what you need** from earlier, **the picture**, the derivations as numbered steps, **numbers and code**, **what to remember**, **common mistakes**, eight **exercises** (six by hand, two in code) and a three-question **self-test**.

How to read it

Lessons 19 and 20 are the summit that the course has been climbing, and well worth reading slowly. If the Girsanov and numéraire lessons (25 and 27) feel abstract the first time, keep two pictures in mind: the paths don't move, only their weights change; and the weights are what shift the bell curve, as in Lesson 20. The numerical lessons reward a different habit: open the notebook, change a number, and see whether the error behaves as the text says.

Several results in this part are surprising and worth checking for yourself rather than taking on trust. The hedge really does work on nearly every path, whichever way the share goes. The third branch of a trinomial tree doesn't make its error smaller. A discounted real-world average really isn't a price.

Notebooks and solutions

Every number, table and figure is computed by a companion Python notebook, one per lesson, using numpy only. The notebooks are free at github.com/RatesWithBip/curriculum.

The solutions are **not** in this book: they are in the companion volume, *Part 4: Solutions*, with every derivation written out and every code solution shown with what it prints. Try each exercise before you look.

Reading alongside

Steven Shreve's *Stochastic Calculus for Finance II*, Chapters 4 to 6 and 9, covers the theory with full rigour; Paul Glasserman's *Monte Carlo Methods in Financial Engineering* and Daniel Duffy's *Finite Difference Methods in Financial Engineering* go much further on the numerics. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and both books are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

LESSON 19 · VIDEO 19

The Black-Scholes Equation

A self-financing portfolio holding Δ shares; Itô's lemma on the option's value; matching the random terms gives the hedge Δ = V_S; matching the rest gives the Black-Scholes equation; and why the share's expected return vanishes from it.

Why this lesson

An option's value depends on the share price and on time: $V(t, S)$. Everything in Part 3 was building towards one question about that function: **what equation must it satisfy?**

The answer is a partial differential equation, found in 1973 by Fischer Black, Myron Scholes and Robert Merton:

$$V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} = rV.$$

The argument behind it is the idea of Lesson 9, moved to continuous time: if you can copy the option with a portfolio of shares and cash, the option must cost what the portfolio costs. Here, Itô's lemma does the matching. Two things come out together: the equation, and the exact recipe for hedging. A third thing comes out as a surprise: the share's expected return is nowhere in the result.

What you need from earlier lessons

- **Replication** (Lesson 9). A portfolio with the same payoff as an option must have the same price, or there is free money.
- **Partial derivatives** (Lesson 2). V_t, V_S, V_{SS} mean: differentiate with respect to t, then S, then S twice, holding the other variable fixed.
- **Itô's lemma** (Lesson 16). For $dX = a\,dt + b\,dW$: $df(t, X) = (f_t + af_x + \tfrac{1}{2}b^2 f_{xx})\,dt + bf_x\,dW$.
- **Geometric Brownian motion** (Lesson 17). $dS = \mu S\,dt + \sigma S\,dW$.

Numbers. From here on, Part 4 uses the course's numbers with a real-world expected return different from r, so that we can watch it disappear: $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$, $\mu = 8\%$. The call is worth 7.1281. (Lesson 20 derives that number.

For now we use the formula as a known function, to check the equation.)

The picture

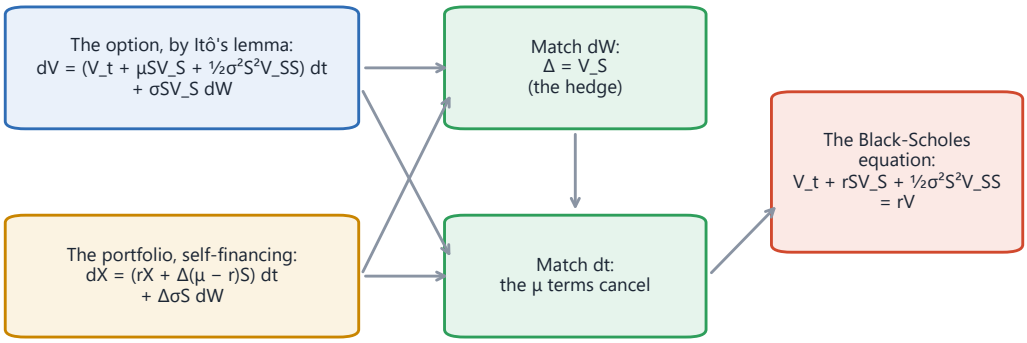


Figure 19.1. The whole derivation on one page. The option's change comes from Itô's lemma, the portfolio's from the self-financing rule. Making them equal has two parts: the dW terms give the hedge, and the dt terms give the equation.

Keep this in view. There are only two ingredients, and one idea: two things that are equal must have equal dW parts and equal dt parts.

A self-financing portfolio

The setup. At time t you hold Δ_t shares. Your total wealth is X_t. What isn't in shares, $X_t - \Delta_t S_t$, sits in cash earning rate r (if negative, you have borrowed at r).

Self-financing means no money enters or leaves after the start. When you change Δ, you sell shares to buy cash or the reverse, and your wealth doesn't jump. So over a short step wealth changes only because the shares moved and the cash earned interest:

$$dX_t = \underbrace{\Delta_t\, dS_t}_{\text{gain on the shares}} + \underbrace{r(X_t - \Delta_t S_t)\, dt}_{\text{interest on the cash}}.$$

A toy example. Wealth 10, with 0.5 shares at 100. So 0.5 × 100 = 50 is in shares, and the cash is $10 - 50 = -40$: you have borrowed 40. Over one trading day, the shares earn an expected $0.5 \times 8\% \times 100/252$, and the loan costs $3\% \times 40/252$. The rule predicts an average change of 0.0111 and a standard deviation of $0.5 \times 20\% \times 100/\sqrt{252} = 0.6299$. In a simulation of 200,000 days the average is 0.0118, the standard deviation 0.6293.

Substituting the share's motion. Put $dS = \mu S\,dt + \sigma S\,dW$ into the rule and collect terms:

$$dX_t = \big(rX_t + \Delta_t(\mu - r)S_t\big)\,dt + \Delta_t \sigma S_t\,dW_t. \tag{1}$$

Read the dt term: wealth earns the cash rate, plus, on the money in shares, the extra return $\mu - r$.

Itô's lemma on the option

Now the option. Let $V(t, S)$ be its value when the share is at S at time t. We don't yet know this function; we assume only that it is smooth enough to differentiate. Itô's lemma with $a = \mu S, b = \sigma S$ says

$$dV = \Big(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\Big)dt + \sigma S V_S\,dW. \tag{2}$$

Checked on the call. At $S = 100$ with a year left, small bumps of the formula give $V_S = 0.5024$, $V_{SS} = 0.01995$ and $V_t =$ -5.2827. Equation (2) then predicts that over one day the option changes on average by 0.0108, with standard deviation 0.6330. The simulated values are 0.0115 and 0.6334.

Matching the dW terms: the hedge

Suppose the portfolio copies the option exactly: $X_t = V(t, S_t)$ for all t. Then $dX = dV$, so (1) and (2) must agree. Two expressions in dt and dW can only agree if their dW parts agree and their dt parts agree, because the dW part is random with mean zero, and the dt part isn't.

Step 1. The dW parts. $\Delta_t \sigma S_t = \sigma S_t V_S$. Divide by σS_t:

$$\boxed{\Delta_t = V_S(t, S_t)}$$

Hold as many shares as the slope of the option's value. This is **delta hedging**.

Why this removes the risk. A small share move δS changes the option by about $V_S\,\delta S$ and the shares by $\Delta\,\delta S$. With $\Delta = V_S$, the two cancel.

Checked. Own the option and short $V_S = 0.5024$ shares. Here is the spread of one change in value, without and with the hedge:

Standard deviation of the change in value over one step, starting at S = 100 with a year left, on 200,000 paths.

DAYS	OPTION ALONE, SD	HEDGED, SD
21	3.0230	0.4770
5	1.4281	0.1120
1	0.6334	0.0223
0.25	0.3162	0.0056

Over one day the hedge cuts the spread by a factor of 28. And notice the last column: it falls in proportion to the step, while the option's own spread falls only like $\sqrt{\text{step}}$. As the step shrinks to nothing, so does the hedged risk. That is what makes the portfolio argument exact in continuous time. (Lesson 24 measures what's left when you can only rebalance so often.)

Matching the dt terms: the equation

Step 2. The dt parts. Using $\Delta = V_S$ and $X = V$ in (1), and comparing with (2):

$$rV + V_S(\mu - r)S = V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}.$$

Step 3. Cancel. On the left is $\mu S V_S - r S V_S$. On the right, $\mu S V_S$. So $\mu S V_S$ goes from both sides, and moving rSV_S across:

$$\boxed{V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} = rV}$$

This is the **Black-Scholes equation**. Together with the payoff at expiry, $V(T, S) = \max(S - K, 0)$ for a call, it fixes the function V for all earlier times.

Reading the equation. The two sides say something simple. The right side, rV, is what the option's value would earn in cash. The left side is what a delta-hedged position actually earns from each source: time decay V_t, interest effects rSV_S, and convexity $\frac{1}{2}\sigma^2 S^2 V_{SS}$. Being riskless, it must earn exactly the cash rate.

In numbers. At $S = 100$ with a year left:

TERM	VALUE
V_t	-5.2827
rSV_S	1.5072
$\frac{1}{2}\sigma^2 S^2 V_{SS}$	3.9893
Sum	0.2138
rV	0.2138

The largest terms are time decay and convexity, and they nearly cancel: this balance between the two is the main subject of Lessons 21 and 24.

Across the grid. At share prices 70 to 130 and times to expiry 0.25 to 1, the largest gap between the two sides is 5.5e-07 for the call and 5.4e-07 for the put, which is the rounding error of the bumps.

Other solutions. The equation is linear, and many functions solve it. Each corresponds to a different payoff.

- $V = S$ holds one share: $V_t = 0$, $V_S = 1$, $V_{SS} = 0$, so the left side is rS, and the right side is rS. ✓
- $V = Ke^{-r(T-t)}$ is cash that will grow to K. Here $V_t = rV$ and the other derivatives vanish. ✓
- $V = S^2 e^{(r+\sigma^2)(T-t)}$ pays S_T^2 at T (Exercise 19.5). The residual is 7.0e-07. Without the σ^2, the residual is 406.

The expected return disappears

Look at the boxed equation again. It has r and σ, but μ cancelled in Step 3. **The option's value doesn't depend on what the share is expected to earn.**

This is the same surprise as in Lesson 9, and the reason is the same. The hedger holds Δ shares against the option, so any drift in the share affects both sides equally. Two investors who disagree completely about μ still agree on the price, because each can hedge the option and remove the very risk they disagree about.

Simulating one day of the hedged position with different μ:

One day, S = 100, one year left. The option's expected change varies strongly with μ. The hedged position's hardly moves.

μ, %	OPTION ALONE, MEAN	HEDGED, MEAN	HEDGED, SD
0	-0.0045	-0.0052	0.0223
8	0.0115	-0.0052	0.0223
20	0.0354	-0.0051	0.0224
40	0.0755	-0.0050	0.0227

The option's average gain goes from -0.0045 to 0.0755. The hedged position stays near -0.0051, which is the cash rate on its value: $r(V - \Delta S)\,dt$. It is negative because the position is worth $V - \Delta S \approx 7.13 - 50.24 = -43.11$: long the option and short half a share, it behaves like a riskless debt, and a debt costs interest.

What's left of μ? Only a tiny effect through the remainder of the hedge, of size $\mu\,dt$ times gamma terms, and it goes to zero as the step shrinks.

What to remember

- **Self-financing:** $dX = \Delta\,dS + r(X - \Delta S)\,dt$.
- **Replicate:** set $X = V$. Compare the two sides with Itô's lemma.
- **The dW terms give the hedge:** $\Delta = V_S$.
- **The dt terms give the equation:** $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$, with the payoff as the final condition.
- μ cancels. The price depends on r and σ, not on the share's expected return.
- A hedged option earns the risk-free rate; the equation is that statement, term by term.

Common mistakes

- **Thinking μ is "in the price somewhere".** It cancels exactly. It does affect how often the option pays off in the real world, but not what it costs.
- **Treating Δ as a constant.** It changes as S and t change. Each short step uses the current V_S; the rebalancing is what the self-financing rule pays for.
- **Forgetting the convexity term.** Ordinary calculus would miss $\frac{1}{2}\sigma^2 S^2 V_{SS}$, and then V would solve the wrong equation.
- **Mixing up V_t with V_τ.** V_t is the change as calendar time goes forward; with $\tau = T - t$, $V_t = -V_\tau$. For a call, V_t is negative.
- **Assuming the equation "knows" the option.** It knows nothing about strikes. Only the final condition tells it which contract you mean.

Exercises

Solutions are in *Part 4: Solutions*.

19.1 A portfolio holds Δ_t shares and the rest in cash at rate r. Derive $dX_t = \Delta_t\,dS_t + r(X_t - \Delta_t S_t)\,dt$, then show with the product rule that $d(e^{-rt}X_t) = \Delta_t\,d(e^{-rt}S_t)$.

19.2 Apply Itô's lemma to $V(t, S_t)$ with $dS = \mu S\,dt + \sigma S\,dW$, writing out each term of the Taylor expansion and showing why the others vanish.

19.3 Derive the Black-Scholes equation a second way: form $\Pi = V - V_S S$, with V_S held fixed over the step, show that $d\Pi$ has no dW term, and argue that Π must earn the risk-free rate.

19.4 Check that $V = S$ and $V = Ke^{-r(T-t)}$ both solve the equation. What contract does their difference describe, and why must it solve the equation too?

19.5 Find the constant b for which $V = S^2 e^{b(T-t)}$ solves the equation. A contract pays S_T^2 at T: what is it worth today with the course's numbers?

19.6 A share pays a continuous dividend yield q: holding it pays $qS\,dt$ in cash. Rewrite the self-financing equation and show that the Black-Scholes equation becomes $V_t + (r-q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$.

19.7 *(Code)* Use small bumps of the formula to check the equation for the call and the put at $S = 70, 85, 100, 115, 130$ and $T - t = 0.25, 0.5, 1$. What is the largest residual? Then try the payoff-shaped function $V = \max(S - K, 0)$ at $S = 100$ and $S = 110$: which terms fail?

19.8 *(Code)* For the call with one year left, simulate the hedged change $V(t+h, S_{t+h}) - V(t, S_t) - V_S(S_{t+h} - S_t)$ with h of 16, 4 and 1 trading days. By what factor does its standard deviation shrink each time h is divided by 4, and why?

Self-test

1. How many shares does the hedge hold, in terms of V?
2. Why does μ not appear in the equation?
3. At $S = 100$ with a year left, which two terms of the equation are largest, and why do they nearly cancel?

Companion notebook: `s5e19_bs_pde.ipynb`. Every number in this chapter is computed by it.

LESSON 20 · VIDEO 20

The Black-Scholes Formula

The call as a discounted risk-neutral average; the exercise condition $Z > -d_-$; splitting the payoff into a share term and a strike term; completing the square to turn a tilted bell curve into a shifted one; the formula, and three independent checks.

Why this lesson

Lesson 19 gave an equation for the option's value. This lesson gives the **answer**:

$$C = S_0 N(d_+) - K e^{-rT} N(d_-), \qquad d_\pm = \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

You have probably seen it before, perhaps as a thing to memorise. Here you will derive it in full, line by line, using nothing more than the normal distribution and one algebra trick. And you will see why there are two terms, each with its own $N(\cdot)$.

There are two routes to the formula: solve the equation from Lesson 19, or compute the discounted risk-neutral average. The second is much easier. Lessons 26 and 28 prove that the two routes must meet; here we take the easier one.

What you need from earlier lessons

- **The pricing rule** (Lesson 11, proved in continuous time in Lesson 26). The price is the discounted average payoff under the risk-neutral probability $\tilde{P}$, where the share drifts at r.
- **The risk-neutral share** (Lessons 5 and 17). $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}$ with $Z \sim N(0,1)$.
- **The normal distribution** (Lesson 4). Density $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$, $N(x) = P(Z \le x)$, and the symmetry $P(Z > -x) = N(x)$.
- **Completing the square** (Lesson 4).

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$.

The picture

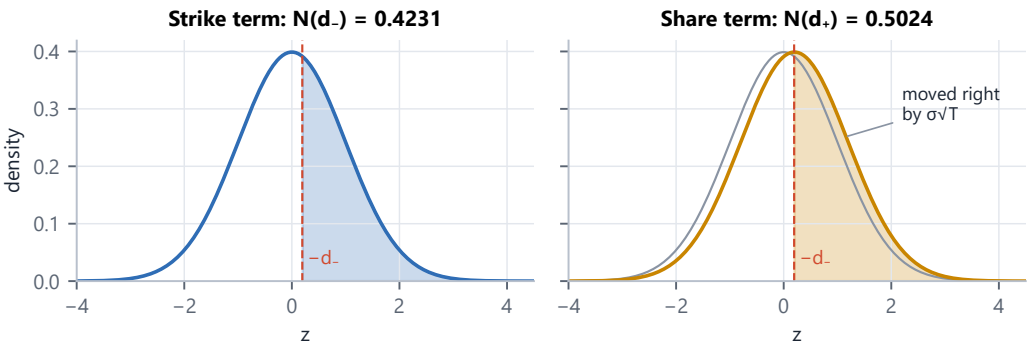


Figure 20.1. The two probabilities in the formula. Both are areas to the right of the same red line. Left: the plain bell curve, with area N(d₋). Right: the bell curve moved right by σ√T, with the larger area N(d₊). Grey is the plain curve for comparison.

The whole derivation is about how the second curve arises.

When is the call exercised?

The call pays only when $S_T > K$. Take logs and solve for Z, one step at a time.

Step 1. $S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z} > K$.

Step 2. Divide by S_0 and take logs: $(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z > \ln(K/S_0)$.

Step 3. Solve for Z, using $\ln(K/S_0) = -\ln(S_0/K)$:

$$Z > \frac{-\ln(S_0/K) - (r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = -d_-, \qquad d_- = \frac{\ln(S_0/K) + (r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

In numbers. $\ln(100/105)$ = -0.04879, the drift is 0.01, the volatility over a year is 0.2. So d_- = -0.19395, and the call is exercised when $Z >$ 0.19395. Check: at that Z, the share ends at exactly 105.00, the strike.

Two terms

The price is the discounted average of the payoff. Since the payoff is $S_T - K$ when $S_T > K$, and zero otherwise, write it with an indicator $\mathbf{1}\{S_T > K\}$ (1 when true, 0 when false), and split it:

$$C = e^{-rT}\tilde{E}\big[(S_T - K)\mathbf{1}\{S_T > K\}\big] = \underbrace{e^{-rT}\tilde{E}\big[S_T\mathbf{1}\{S_T > K\}\big]}_{\text{the share term}} - \underbrace{Ke^{-rT}\tilde{P}(S_T > K)}_{\text{the strike term}}.$$

In words: **receive** the share if it ends above K, and **pay** K if it ends above K.

We estimate both from 2,000,000 simulated prices before computing them exactly: the share term is about 50.232, the strike term about 43.103.

The strike term

This is the easy one. Using the exercise condition and the symmetry of the normal:

$$\tilde{P}(S_T > K) = P(Z > -d_-) = P(Z < d_-) = N(d_-).$$

So the strike term is $Ke^{-rT}N(d_-)$. With our numbers $N(d_-) = 0.42311$ and $Ke^{-rT} = 101.8968$, which multiply to 43.1133.

$N(d_-)$ is the risk-neutral chance of finishing in the money, about 42%.

The share term: completing the square

Now the harder one. Write the average as an integral against the normal density, over the values of z where the call is exercised.

Step 1. The integral.

$$e^{-rT}\tilde{E}\big[S_T \mathbf{1}\{S_T > K\}\big] = e^{-rT}\int_{-d_-}^{\infty} S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}z}\,\varphi(z)\,dz.$$

Step 2. Tidy the constants. $e^{-rT}e^{rT}$ cancels. What's left outside is $S_0 e^{-\frac{1}{2}\sigma^2 T}$:

$$= S_0 e^{-\frac{1}{2}\sigma^2 T}\int_{-d_-}^{\infty} e^{\sigma\sqrt{T}z}\varphi(z)\,dz.$$

Step 3. Combine the exponents. Write $a = \sigma\sqrt{T}$ for short. Inside the integral:

$$e^{az}\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{az-z^2/2}.$$

Step 4. Complete the square. $-\frac{1}{2}z^2 + az = -\frac{1}{2}(z-a)^2 + \frac{1}{2}a^2$. (Expand the right side to check.) So

$$e^{az}\varphi(z) = e^{\frac{1}{2}a^2}\frac{1}{\sqrt{2\pi}}e^{-(z-a)^2/2} = e^{\frac{1}{2}a^2}\varphi(z-a).$$

This is the heart of the derivation. Multiplying a bell curve by an exponential doesn't change its shape: it moves it to the right by a, and scales it by a constant. That's the second curve in the picture.

Step 5. The constants cancel. With $a^2 = \sigma^2 T$, the constant $e^{\frac{1}{2}\sigma^2 T}$ exactly cancels the $e^{-\frac{1}{2}\sigma^2 T}$ outside:

$$= S_0 \int_{-d_-}^{\infty} \varphi(z - \sigma\sqrt{T})\, dz.$$

Step 6. Substitute $u = z - \sigma\sqrt{T}$. The lower limit becomes $-d_- - \sigma\sqrt{T}$:

$$= S_0 \int_{-d_- - \sigma\sqrt{T}}^{\infty} \varphi(u)\, du = S_0 N(d_- + \sigma\sqrt{T}) = S_0 N(d_+).$$

Here $d_+ = d_- + \sigma\sqrt{T}$, which is the same formula with $+\frac{1}{2}\sigma^2$ in place of $-\frac{1}{2}\sigma^2$.

In numbers. $d_+ = 0.00605$, and $N(d_+) = 0.50241$, so the share term is 50.2413. A direct numerical integral of Step 1 gives 50.2413.

The formula, checked three ways

Put the two terms together:

$$\boxed{C = S_0 N(d_+) - K e^{-rT} N(d_-)}$$

$$= 50.2413 - 43.1133 = 7.1281.$$

Check 1. A direct integral of payoff times density: 7.1281.

Check 2. Monte Carlo with 2,000,000 paths: 7.1285, with a standard error of 0.0089.

Check 3. The equation. Lesson 19 checked that the formula satisfies the Black-Scholes equation, and at expiry it is the payoff. So it is the solution.

The put. The same steps, or put-call parity $P = C - S_0 + Ke^{-rT}$, give $P = Ke^{-rT} N(-d_-) - S_0 N(-d_+) = 9.0248$.

Other strikes. Monte Carlo below uses one set of paths for all strikes:

S_0 = 100, r = 3%, σ = 20%, T = 1. Simulation on 2,000,000 paths.

STRIKE	d_+	d_-	CALL	CALL, SIMULATED	PUT
85	1.0626	0.8626	19.1313	19.1315	1.6192
95	0.5065	0.3065	12.1797	12.1795	4.3720
105	0.0060	-0.1940	7.1281	7.1285	9.0248
115	-0.4488	-0.6488	3.8596	3.8600	15.4608
125	-0.8657	-1.0657	1.9521	1.9531	23.2578

What μ did. Nothing: it isn't in the formula. Every path in this lesson drifts at r, not at μ. That isn't because the world behaves so, but because that is the average that gives the price.

A way to read the formula. $N(d_+)$ is the delta (Lesson 21), so $S_0 N(d_+)$ is the money in shares in the hedge, and $Ke^{-rT}N(d_-)$ is what's borrowed. **The formula is the hedge portfolio.** With our numbers: about half a share (worth 50.24), financed by a loan of 43.11, for a net cost of 7.13.

What to remember

- **Exercise condition:** $S_T > K \iff Z > -d_-$.
- **Split:** $C = e^{-rT}\tilde{E}[S_T \mathbf{1}\{S_T > K\}] - Ke^{-rT}\tilde{P}(S_T > K)$.
- **Strike term:** $Ke^{-rT}N(d_-)$. $N(d_-)$ is the risk-neutral probability of exercise.
- **Share term:** $S_0 N(d_+)$, by completing the square. The exponential moves the bell curve right by $\sigma\sqrt{T}$.
- $d_+ = d_- + \sigma\sqrt{T}$.
- **Numbers:** $d_+ = 0.0060$, d_- = -0.1940, $C = 7.1281$, $P = 9.0248$.

Common mistakes

- **Reading $N(d_+)$ as the chance of exercise.** Under $\tilde{P}$ that's $N(d_-)$. $N(d_+)$ is a chance under a different probability (Lesson 27).
- **Using μ in $d_\pm$.** They use r. Use μ, and you get the average payoff of a real-world bet, not a price.
- **Using σ where $\sigma\sqrt{T}$ belongs.** Both σ and T are in the denominator of $d_\pm$; they only look the same here because $T = 1$.
- **Sign slips on $\ln(S_0/K)$.** It's negative when the strike is above the share.
- **Forgetting to discount the strike.** Ke^{-rT}, not K.

Exercises

Solutions are in *Part 4: Solutions*.

20.1 Starting from $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z}$, show that $S_T > K$ exactly when $Z > -d_-$, and evaluate d_- and the critical Z with the course's numbers.

20.2 Given $N(0.00605) = 0.50241$ and $N(-0.19395) = 0.42311$, compute the call price by hand, and say what each of the two terms is worth.

20.3 Compute $e^{-rT}\tilde{E}[S_T \mathbf{1}\{S_T > K\}]$ in full: write the integral, combine the exponents, complete the square, substitute, and show that it equals $S_0 N(d_+)$.

20.4 Derive the put formula $P = Ke^{-rT}N(-d_-) - S_0 N(-d_+)$ by the same steps, and check that it agrees with put-call parity.

20.5 A **cash digital** pays 1 if $S_T > K$; an **asset digital** pays S_T if $S_T > K$. Write down their prices. Show that a call is an asset digital minus K cash digitals, and evaluate all three.

20.6 Find the limit of the call formula (a) as $\sigma \to 0$, (b) as $T \to 0$, (c) as $S_0 \to \infty$, and explain each in words.

20.7 *(Code)* Price the call for strikes 85, 95, 105, 115, 125 by numerically integrating payoff times density (trapezium rule), and compare with the formula. How fine must the grid be for four decimals?

20.8 *(Code)* Estimate the cash digital and the call by Monte Carlo with 10^6 paths, with 95% confidence intervals. Which has the smaller error relative to its price, and why?

Self-test

1. What is the risk-neutral probability that the call is exercised, with the course's numbers?
2. What does the exponential $e^{\sigma\sqrt{T}z}$ do to the bell curve?
3. What are the two terms of the formula worth with the course's numbers, and what portfolio do they describe?

Companion notebook: `s5e20_bs_formula.ipynb`. Every number in this chapter is computed by it.

LESSON 21 · VIDEO 21

The Greeks

The identity Sφ(d₊) = Ke^(−rτ)φ(d₋), which makes the derivatives clean; delta, gamma and theta derived step by step; the balance between theta and gamma that the Black-Scholes equation states; vega; and how all of them look across share prices and as expiry nears.

Why this lesson

A price is one number. A trader needs to know how that number moves. The **Greeks** are the partial derivatives of the option's value, each named after a letter:

GREEK	DEFINITION	ANSWERS
Delta, Δ	$\partial V/\partial S$	How many shares hedge the option?
Gamma, Γ	$\partial^2 V/\partial S^2$	How fast does the hedge go out of date?
Theta, Θ	$\partial V/\partial t$	What does a day's waiting cost?
Vega, $\mathcal{V}$	$\partial V/\partial \sigma$	What if volatility is wrong?

(Vega isn't a Greek letter. The name stuck anyway.)

Differentiating the formula looks like a mess, because d_+ and d_- depend on S and t. But one identity makes the mess collapse. That identity is the first thing to learn.

What you need from earlier lessons

- **The formula** (Lesson 20). $C = SN(d_+) - Ke^{-r\tau}N(d_-)$, with $\tau = T - t$, $d_\pm = \dfrac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$, $d_+ = d_- + \sigma\sqrt{\tau}$.
- **The equation** (Lesson 19). $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$.
- **The chain rule** (Lesson 1). $\frac{d}{dx}N(g(x)) = \varphi(g(x))\,g'(x)$, because $N' = \varphi$.

Numbers. $S = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $\tau = 1$.

The picture

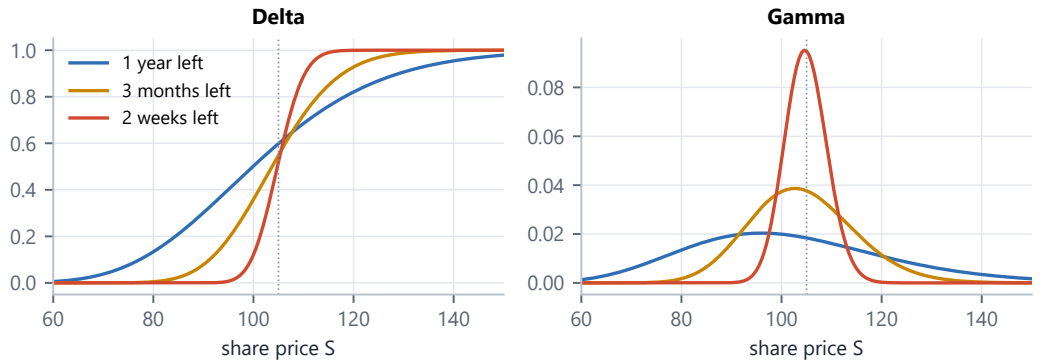


Figure 21.1. Delta and gamma of the call against the share price, with one year, three months and two weeks left. The dotted line is the strike. Near expiry, delta becomes almost a step, and gamma a tall narrow spike.

Gamma is the slope of delta. Where delta is steep, the hedge goes out of date quickly, and that happens near the strike, close to expiry.

An identity that does the work

$$\boxed{S\varphi(d_+) = Ke^{-r\tau}\varphi(d_-)}$$

Proof. Step 1. Since $d_+ = d_- + \sigma\sqrt{\tau}$, expand the square: $d_+^2 = d_-^2 + 2d_-\sigma\sqrt{\tau} + \sigma^2\tau$.

Step 2. So $\varphi(d_+) = \varphi(d_-)e^{-\frac{1}{2}(2d_-\sigma\sqrt{\tau}+\sigma^2\tau)} = \varphi(d_-)e^{-d_-\sigma\sqrt{\tau}-\frac{1}{2}\sigma^2\tau}$.

Step 3. By the definition of d_-: $d_-\sigma\sqrt{\tau} = \ln(S/K) + r\tau - \frac{1}{2}\sigma^2\tau$. Substituting, the $\frac{1}{2}\sigma^2\tau$ cancels:

$$\varphi(d_+) = \varphi(d_-)e^{-\ln(S/K)-r\tau} = \varphi(d_-)\frac{K}{S}e^{-r\tau}.$$

Step 4. Multiply by S. ■

Checked. At $S = 100$, $\tau = 1$: left 39.893498, right 39.893498. The largest gap across four cases is 7.1e-15.

Delta

Differentiate C with respect to S. The product rule on the first term gives two pieces, and the chain rule handles $N(d_\pm)$. With $\partial d_\pm/\partial S = 1/(S\sigma\sqrt{\tau})$, the same for both:

$$\frac{\partial C}{\partial S} = N(d_+) + \underbrace{S\varphi(d_+)\frac{1}{S\sigma\sqrt{\tau}} - Ke^{-r\tau}\varphi(d_-)\frac{1}{S\sigma\sqrt{\tau}}}_{\text{zero, by the identity}}.$$

$$\boxed{\Delta = N(d_+)}$$

In numbers. $\Delta = 0.5024$. A small bump of the formula gives 0.5024. The put's delta, from put-call parity, is $N(d_+) - 1 =$ -0.4976.

Meaning. About half a share hedges this call. Delta runs from 0 (far below the strike: the option barely reacts) to 1 (far above: it moves one-for-one with the share).

Gamma

Differentiate $\Delta = N(d_+)$ again, with the chain rule:

$$\boxed{\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\varphi(d_+)}{S\sigma\sqrt{\tau}}}$$

In numbers. $\Gamma = 0.01995$ (bump: 0.01995). So if the share rises by 1, delta rises by about 0.020, from 0.502 to about 0.522.

Near expiry. The $\sqrt{\tau}$ in the denominator is why gamma explodes. At the strike:

Gamma at S = K grows like 1/√τ: a quarter of the time left means about twice the gamma.

Time left τ	GAMMA AT S = K
1.00	0.0184
0.25	0.0377
0.04	0.0949
0.01	0.1899

Gamma is the same for a call and a put, because their difference, $S - Ke^{-r\tau}$, is a straight line in S.

Theta, and its balance with gamma

Differentiate with respect to t. Now $\tau = T - t$, so $\partial\tau/\partial t = -1$. Direct differentiation of the formula, after the identity cancels the $\partial d_\pm/\partial t$ pieces (Exercise 21.4), leaves two terms:

$$\boxed{\Theta = -\frac{S\varphi(d_+)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_-)}$$

In numbers. Θ = -5.2827 per year (bump: -5.2827), or -0.0210 per trading day. The first term is -3.9893, the second -1.2934. Most of the decay comes from volatility running out.

The balance. Lesson 19's equation, in terms of Greeks, is

$$\Theta + rS\Delta + \tfrac{1}{2}\sigma^2 S^2 \Gamma = rV.$$

The Black-Scholes equation term by term, at τ = 1. Each row adds up: the first three columns sum to the last.

S	Θ	rSΔ	½σ²S²Γ	RV
80	-2.0152	0.3206	1.7243	0.0297
100	-5.2827	1.5072	3.9893	0.2138
105	-5.6494	1.8859	4.0600	0.2965
120	-5.4762	2.9542	3.1422	0.6202

Look at the columns for Θ and $\frac{1}{2}\sigma^2 S^2 \Gamma$: they are the big ones, with opposite signs. This is the central trade-off of option trading.

What a hedged holder lives on. Own the option and hedge the delta. Over a short time δt, a Taylor expansion with the balance above gives the profit (Exercise 21.8):

$$\text{profit} \approx \tfrac{1}{2}\Gamma S^2 \left(\left(\tfrac{\delta S}{S}\right)^2 - \sigma^2 \delta t \right).$$

Big moves pay. Quiet days cost. The two balance at a move of $|\delta S/S| = \sigma\sqrt{\delta t}$. Over one day that's 1.26%, or 1.26 on a share at 100.

Vega

The derivative with respect to volatility, again with the identity cancelling the messy terms (Exercise 21.6):

$$\mathcal{V} = \frac{\partial C}{\partial \sigma} = S\varphi(d_+)\sqrt{\tau} = \sigma\tau S^2 \Gamma.$$

In numbers. $\mathcal{V}$ = 39.89 (bump: 39.89). One point of volatility is 0.01, so a one-point move changes the price by about 0.40. The call is worth 7.13, so a one-point error in volatility is a 5.6% error in price.

The Greeks across prices

The call's Greeks at τ = 1.

S	CALL	Δ	Γ	Θ	VEGA
80	0.990	0.134	0.0135	-2.015	17.24
90	3.114	0.301	0.0194	-3.855	31.35
100	7.128	0.502	0.0199	-5.283	39.89
105	9.884	0.599	0.0184	-5.649	40.60
110	13.099	0.685	0.0161	-5.775	39.06
120	20.672	0.821	0.0109	-5.476	31.42
130	29.344	0.906	0.0064	-4.830	21.76

Delta grows steadily with S. Gamma and vega are both built from $\varphi(d_+)$, which is largest where $d_+ = 0$, close to $S = 100$ here. The extra factor of $1/S$ in gamma pulls its peak a little lower, and the factor of S in vega pulls its peak a little higher.

What to remember

- **The identity:** $S\varphi(d_+) = Ke^{-r\tau}\varphi(d_-)$ cancels the messy terms.
- **Delta:** $N(d_+)$. **Gamma:** $\varphi(d_+)/(S\sigma\sqrt{\tau})$. **Vega:** $S\varphi(d_+)\sqrt{\tau}$.
- **Theta:** $-\dfrac{S\varphi(d_+)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_-)$.
- **The balance:** $\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2\Gamma = rV$. Time decay pays for gamma.
- A hedged holder's day: $\frac{1}{2}\Gamma S^2((\delta S/S)^2 - \sigma^2\delta t)$; break-even move $\sigma\sqrt{\delta t}$.
- Gamma and vega peak near the strike; gamma explodes at the strike near expiry.

Common mistakes

- **Differentiating only the visible S.** $d_\pm$ also depend on S. Delta is $N(d_+)$ only because the extra pieces cancel, by the identity. They are not simply ignored.
- **Thinking delta is the probability of exercise.** It's $N(d_+)$. The risk-neutral chance of exercise is $N(d_-)$.
- **Sign of theta.** With $\Theta = \partial V/\partial t$ (calendar time forward), a call's theta is negative. Many trading systems quote it per day; ours is per year.
- **Bumping too small.** Rounding errors take over: a gamma bump of 10^{-4} is worse than one of 10^{-2} (Exercise 21.7).

- **"Gamma is the risk."** It is the risk **and** the income. A long-gamma position earns on big moves, and pays theta for the privilege.

Exercises

Solutions are in *Part 4: Solutions*.

21.1 Prove that $S\varphi(d_+) = Ke^{-r\tau}\varphi(d_-)$. (Write $d_+ = d_- + \sigma\sqrt{\tau}$, expand d_+^2, and use the definition of d_-.)

21.2 Differentiate $C = SN(d_+) - Ke^{-r\tau}N(d_-)$ with respect to S, using the chain rule on each term, and use the identity to show $\Delta = N(d_+)$.

21.3 Differentiate Δ to find $\Gamma = \varphi(d_+)/(S\sigma\sqrt{\tau})$. Evaluate it at $S = 100$, $\tau = 1$. If the share rises by 1, roughly how much does delta change?

21.4 Differentiate C with respect to t to find Θ, and verify the Black-Scholes equation directly from the formulas for Θ, Δ and Γ.

21.5 Use put-call parity to find the put's delta, gamma and theta from the call's. Which of them are the same as the call's, and why?

21.6 Show that **vega**, $\partial C/\partial\sigma$, equals $S\varphi(d_+)\sqrt{\tau}$, and that it equals $\sigma\tau S^2\Gamma$. Why are vega and gamma so closely related?

21.7 *(Code)* Compute delta, gamma, theta and vega at $S = 80, 100, 120$ by small bumps of the formula and compare with the formulas. How does the error of the gamma bump change as the bump size runs through $1, 0.1, 0.01, 0.001, 0.0001$?

21.8 *(Code)* Buy the call at $S = 100$, sell Δ shares, finance the difference at rate r, and let one trading day pass with moves of 0, $\pm 0.5\%$, $\pm 1.26\%$ and $\pm 2.5\%$. Compare the exact one-day profit with $\frac{1}{2}\Gamma S^2\big((\delta S/S)^2 - \sigma^2\delta t\big)$. Where is the break-even move?

Self-test

1. What are the call's delta and gamma at $S = 100$, $\tau = 1$?
2. Why is gamma the same for a call and a put?
3. A hedged option holder has a quiet day, with the share unchanged. Do they make or lose money, and which Greek says so?

Companion notebook: `s5e21_greeks.ipynb`. Every number in this chapter is computed by it.

LESSON 22 · VIDEO 22

Checking the Formula Numerically

Monte Carlo and its standard error, and whether the error bar tells the truth; the binomial tree's convergence and why it jumps up and down; two cures for the jumps; and the American put, where the formula stops and the tree carries on.

Why this lesson

A formula you can't check is a formula you can't trust. Two numerical methods should reach the same answer as Black-Scholes by entirely different roads:

- **Monte Carlo** averages simulated payoffs (the expectation of Lesson 20).
- **The binomial tree** works backwards through a lattice (the replication of Part 2).

But "should reach" hides the real questions. How close is it after n steps or M paths? How do you know, on a problem with no formula to compare against? This lesson answers those, and finishes with the first problem the formula can't do: the American put.

What you need from earlier lessons

- **The central limit theorem** (Lesson 6). The average of M independent draws is approximately normal, with standard deviation $s/\sqrt{M}$.
- **The binomial tree** (Lessons 9 to 12). Value = discounted average of the next two nodes; with $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, and $\tilde{p} = (e^{r\Delta t} - d)/(u - d)$, the Cox-Ross-Rubinstein tree.
- **The American put on a tree** (Lesson 12). At each node, take the larger of holding and exercising.
- **The formula** (Lesson 20). Call 7.1281, put 9.0248.

The picture

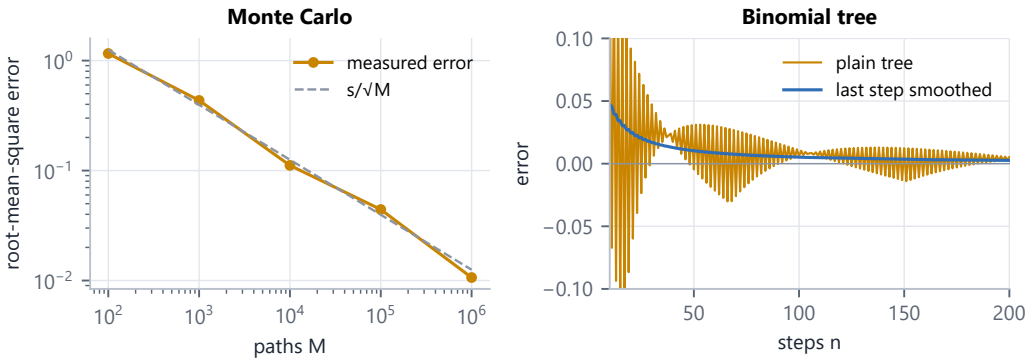


Figure 22.1. Left: the size of Monte Carlo's error against the number of paths, on log scales. It follows s/√M closely, without any wiggles, but falls slowly. Right: the error of a binomial tree against its number of steps. It jumps up and down (amber). Smoothing the last step (blue) removes the jumps and leaves a steady, shrinking error.

Monte Carlo has a slow but tidy error. A tree has a small but untidy one. Each method's weakness can be measured, and to some extent cured.

Monte Carlo and its error bar

The method. Draw $Z_1, \dots, Z_M$ independent standard normals. Compute the discounted payoff of each path, $Y_i = e^{-rT}\max(S_T(Z_i) - K, 0)$. The estimate is the sample mean $\bar{Y}$.

The error bar. $\bar{Y}$ is random, since another set of draws gives another answer. Its standard deviation, the **standard error**, is

$$\mathrm{SE} = \frac{s}{\sqrt{M}},$$

where s is the standard deviation of one Y_i, itself estimated from the sample. By the central limit theorem, $\bar{Y}$ is nearly normal, so $\bar{Y} \pm 1.96\,\mathrm{SE}$ is an approximate **95% confidence interval**.

The numbers. Here $s = 12.54$.

Monte Carlo for the call. The formula gives 7.1281.

PATHS M	ESTIMATE	STANDARD ERROR	95% LOW	95% HIGH	TRUE ERROR
1,000	7.5666	0.4173	6.7486	8.3846	0.4385
10,000	7.0591	0.1226	6.8188	7.2993	-0.0690
100,000	7.0960	0.0394	7.0186	7.1733	-0.0321
1,000,000	7.1314	0.0126	7.1068	7.1560	0.0034
10,000,000	7.1294	0.0040	7.1216	7.1371	0.0013

Reading the table. Each 100-fold increase in paths gains one more digit. That's the $1/\sqrt{M}$: to halve the error you need 4 times the paths. For an error bar of ± 0.01, you need about $(1.96 \times 12.54/0.01)^2 \approx 6,039,784$ paths.

The great virtue of this rate: **it doesn't depend on the number of assets**. Lesson 30 comes back to this.

Does the error bar tell the truth?

The claim "95%" can be tested. Run the whole experiment 2,000 times, with 10,000 paths each time, and count how often the interval contains the true price.

- **Coverage:** 0.947, very close to 0.95.
- **Spread of the 2,000 estimates:** 0.1249. Average of the reported standard errors: 0.1254. Each run's own error bar correctly reports how far off that run is likely to be.
- **Bias:** the average estimate is off by 0.0029, well within noise.

Why this matters. On a problem with no formula, you still get an honest error bar, from the simulation itself. Few numerical methods give you this. The tree below doesn't.

The binomial tree converges, unevenly

Build a tree with n steps, $u = e^{\sigma\sqrt{T/n}}$, and work backwards. As n grows, the price goes to the formula. But:

The binomial tree for the call. One more step can change the error's size and even its sign.

STEPS N	TREE PRICE	ERROR
10	7.29337	0.16530
11	7.06085	-0.06721
50	7.12345	-0.00462
51	7.15916	0.03109
100	7.13399	0.00593
101	7.13847	0.01040
500	7.13013	0.00207
501	7.12944	0.00138
1,000	7.12789	-0.00018
1,001	7.12960	0.00153

Why it jumps. A call's payoff has a kink at K. The tree's final nodes are spaced apart, and K falls somewhere between two of them. As n changes by one, the nodes move, and K lands in a different place relative to them. The tree effectively rounds the kink differently each time.

So one tree size can be lucky: $n = 1000$ is off by -0.00018, while $n = 1001$ is off by 0.00153. **A single size proves nothing.** Judge the method by the worst error over a range of sizes.

Taming the oscillation

Two simple cures.

Average two neighbours. The errors at n and $n + 1$ tend to have opposite signs, so average the two prices.

Smooth the last step. The trouble comes from the kink, which is only in the final payoff. So don't put it on the grid. At the nodes one step before expiry, use the Black-Scholes value for that last short step. By then the value is a smooth function, and the rest of the tree works on something without a kink. (For an option with no formula, a similar smoothing can be done by numerical integration over the last step.)

The worst error over 20 consecutive tree sizes, for the three versions.

FROM N, 20 SIZES	PLAIN TREE	AVERAGED	SMOOTHED
25	0.06292	0.02687	0.02088
50	0.03117	0.01324	0.01049
100	0.01153	0.00863	0.00524
200	0.00568	0.00432	0.00261
400	0.00497	0.00067	0.00133
800	0.00243	0.00026	0.00067

Reading the table. Both cures reduce the worst error. Smoothing has a further virtue: its error halves steadily as n doubles (ratio 1.99 on the last two rows). A steady error can be removed.

Richardson extrapolation. Suppose the error is c/n, with c unknown. Then $V_n \approx V + c/n$ and $V_{2n} \approx V + c/(2n)$. Multiply the second by 2 and subtract the first: the c terms cancel:

$$V \approx 2V_{2n} - V_n.$$

You pay for two runs and get an extra order of accuracy, provided that the error really is steady. On a jumpy sequence, it fails. (Exercises 22.4 and 22.8.)

The American put: what the formula can't do

An American option can be exercised at any time. No closed formula is known for the put. But the tree needs only one extra line: at each node, take the larger of holding and exercising:

$$V_{\text{node}} = \max\left(\text{discounted average of children},\ K - S_{\text{node}}\right).$$

Using neighbour averaging:

The tree's European put (formula 9.0248) and American put.

STEPS N	EUROPEAN	AMERICAN	PREMIUM
50	9.0381	9.4928	0.4547
100	9.0330	9.4872	0.4542
200	9.0292	9.4825	0.4534
400	9.0255	9.4788	0.4533
800	9.0251	9.4780	0.4529
1,600	9.0252	9.4778	0.4526

The American put settles at about 9.478, worth 0.453 more than the European one. And there is a useful check built in: the tree's European column converges to the formula, so the tree can be trusted before you use it on the American column, which has no formula. This is how numerical methods earn their keep: **validate on a case you can check, then use on one you can't.**

What to remember

- **Monte Carlo:** SE $= s/\sqrt{M}$. The interval $\bar{Y} \pm 1.96$ SE really does cover the truth 95% of the time, and doesn't need a formula.
- A digit costs 100 times the paths, whatever the number of assets.
- **Tree:** the error is about $1/n$, but jumps as the strike moves relative to the nodes. One size proves nothing.
- **Cures:** average neighbours; smooth the last step. **Richardson:** $V \approx 2V_{2n} - V_n$, when the error is steady.
- The American put is one line on a tree: about 9.478 here.
- Validate on a case with a formula. Then trust the method on a case without.

Common mistakes

- **Quoting a Monte Carlo price without its error bar.** It tells the reader nothing about accuracy.
- **Trusting one tree size.** A small error at one n may be luck.
- **Using Richardson on a jumpy sequence.** It amplifies the noise.
- **Comparing an estimate with the "true value" on a problem where there isn't one.** Use the standard error, or a finer grid.
- **Testing only on easy cases.** Validate where the method will be stressed: deep out of the money, short expiry.

Exercises

Solutions are in *Part 4: Solutions*.

22.1 One discounted call payoff has a standard deviation of about 12.5. How many paths do you need for a 95% confidence half-width of 0.01? Of 0.001? Why is the second answer 100 times the first, not 10?

22.2 Antithetic variates. Estimate with pairs Z and $-Z$: $\hat{V} = \frac{1}{2}\big(f(Z) + f(-Z)\big)$. Show that its variance is $\frac{1}{2}\mathrm{Var}(f(Z))(1+\rho)$, where ρ is the correlation between $f(Z)$ and $f(-Z)$. Why does this help for a monotone payoff?

22.3 Explain why a call priced on a binomial tree jumps up and down as the steps increase, by looking at where the strike falls among the final nodes. Why is a put on the same tree no better, and why does a smooth payoff not have the problem?

22.4 Richardson extrapolation. If a method's error is c/n for some unknown c, show that $2V_{2n} - V_n$ cancels it. With $V_{100} = 9.0330$ and $V_{200} = 9.0292$ (averaged-neighbour prices of the European put), what is the extrapolated value, and how does it compare with the formula's 9.0248?

22.5 Control variate. You know $E[e^{-rT}S_T] = S_0$ exactly. Show that $\hat{V} = \bar{Y} - b(\bar{X} - S_0)$, with Y the discounted payoff and $X = e^{-rT}S_T$, is unbiased for any b; that the best b is $\mathrm{Cov}(Y, X)/\mathrm{Var}(X)$; and that the variance falls by the factor $1 - \rho^2$.

22.6 A cash digital pays 1 if $S_T > K$, with $\tilde{P}(S_T > K) = 0.4231$. What is the standard deviation of one discounted payoff? How many paths give a 95% half-width of 0.001, and how does the answer change for a strike so high that the probability is 0.01?

22.7 *(Code)* Add antithetic variates and then a control variate to the call's Monte Carlo with 10^5 paths. Report each estimator's standard error, and the number of plain paths each one is worth. Does the control variate help more for $K = 85$ or $K = 125$, and why?

22.8 *(Code)* Price the American put with n of 50, 100, 200, 400, 800 steps, using neighbour averaging, then Richardson extrapolation. Take $n = 6400$ as the benchmark: how close does each method get at each n? Find the share price below which immediate exercise is best with one year left.

Self-test

1. To cut a Monte Carlo error by a factor of 10, how many times more paths do you need?

2. Why isn't a tree's error at a single size a good measure of its accuracy?
3. What does the tree do differently at each node for an American put?

Companion notebook: `s5e22_numerical_checks.ipynb`. Every number in this chapter is computed by it.

LESSON 23 · VIDEO 23

The Trinomial Tree

Why work in log price; three branches and three moment equations; the stretch factor λ, why √3 matches the fourth moment and how λ = 1 falls back to a binomial tree; a small tree by hand; and what the third branch really buys you, which turns out to be steadiness, not size of error.

Why this lesson

A binomial tree has two branches per step, and only just enough freedom to match the mean and the variance of the share. Add a third branch, and you gain one more free number. What can you do with it?

Banks and textbooks use trinomial trees widely, partly for what this lesson finds, and partly because a trinomial tree is the same thing as the explicit finite-difference scheme of Lesson 29. So this lesson is also the first step to grids. We derive the probabilities by matching moments, price by hand, and then compare with the binomial tree honestly: the answer isn't quite what you'd guess.

What you need from earlier lessons

- **The log price** (Lesson 17). Under the risk-neutral probability, $d\ln S = \nu\,dt + \sigma\,d\tilde{W}$, with constant $\nu = r - \frac{1}{2}\sigma^2$.
- **Moments of the normal** (Lesson 4). A normal with mean 0 and variance s^2 has $E[X^4] = 3s^4$.
- **Tree pricing** (Lessons 10 and 22). Discounted average of the children; judge by the worst error over several sizes; Richardson extrapolation.

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$, so $\nu = 0.01$. Call 7.1281, put 9.0248.

The picture

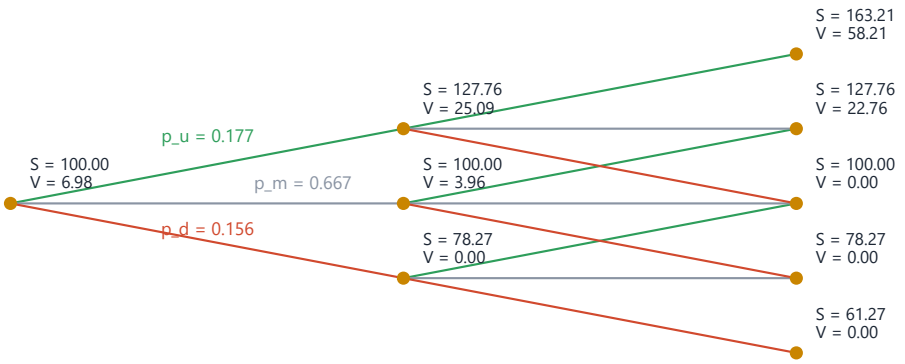


Figure 23.1. A two-step trinomial tree for the call with λ = √3. Each node splits three ways: up, middle, down. Values are worked back from the payoff at the right.

The branches all come back together: up then down lands in the same place as middle then middle. So the tree grows by two nodes per step, not by three times.

Why three branches, and why logs

Why logs. In $X = \ln S$, each step has the same mean, $\nu\Delta t$, and variance, $\sigma^2\Delta t$, wherever you are. So one set of probabilities and one jump size serve the whole tree, and the moves $+\Delta x$, 0, $-\Delta x$ recombine neatly. With $n = 100$ steps a year, $\Delta t = 0.01$, so the target mean per step is 0.0001, and the target standard deviation is 0.02.

Why three. Two branches give two equations' worth of freedom. Three branches, with a jump size, let us match more, or match the same with room left over to steer the tree.

Matching the moments

Let the jump size be Δx, and the probabilities p_u, p_m, p_d. Then one step ΔX has:

- **Total probability:** $p_u + p_m + p_d = 1$.
- **Mean:** $E[\Delta X] = (p_u - p_d)\Delta x$. Set this to $\nu\Delta t$.
- **Variance:** $E[\Delta X^2] = (p_u + p_d)\Delta x^2$. Set this to $\sigma^2\Delta t$, neglecting the very small $(\nu\Delta t)^2$.

Three equations, three unknowns, once Δx is fixed. It's natural to measure Δx against the step's standard deviation:

$$\Delta x = \lambda\sigma\sqrt{\Delta t},$$

with λ a **stretch factor** of our choosing.

Step 1. The variance equation. $p_u + p_d = \dfrac{\sigma^2\Delta t}{\lambda^2\sigma^2\Delta t} = \dfrac{1}{\lambda^2}$.

Step 2. The total. $p_m = 1 - \dfrac{1}{\lambda^2}$.

Step 3. The mean equation. $p_u - p_d = \dfrac{\nu\Delta t}{\Delta x}$.

Step 4. Add and subtract Steps 1 and 3:

$$\boxed{p_u = \frac{1}{2}\left(\frac{1}{\lambda^2} + \frac{\nu\Delta t}{\Delta x}\right), \qquad p_m = 1 - \frac{1}{\lambda^2}, \qquad p_d = \frac{1}{2}\left(\frac{1}{\lambda^2} - \frac{\nu\Delta t}{\Delta x}\right)}$$

In numbers, with $\lambda = \sqrt{3}$ and $n = 100$: $\Delta x = 0.03464$, $p_u = 0.16811$, $p_m = 0.66667$, $p_d = 0.16522$. Check: they sum to 1.0, the mean is 0.000100 (target 0.000100), and the variance is 0.000400 (target 0.000400).

A further check. The share itself should grow at r: $p_u e^{\Delta x} + p_m + p_d e^{-\Delta x} = 1.000300$, and $e^{r\Delta t} = 1.000300$. That's not exact, since we matched log moments, not this one, but it is very close.

The stretch factor

λ is free. Which should we choose?

The fourth moment. On the tree, $E[\Delta X^4] = (p_u + p_d)\Delta x^4 = \dfrac{1}{\lambda^2}\lambda^4\sigma^4\Delta t^2 = \lambda^2\sigma^4\Delta t^2$. A normal step has $3\sigma^4\Delta t^2$. So $\lambda = \sqrt{3}$ **matches the fourth moment too**: with our numbers, 4.8e-07 against 4.8e-07.

The third moment comes along too (Exercise 23.2). So $\lambda = \sqrt{3}$ is the natural default, with the pleasing probabilities $\frac{1}{6}, \frac{2}{3}, \frac{1}{6}$ before the small tilt.

The choice of λ, with n = 100. A normal has fourth-moment coefficient 3.

λ	Δx	P_U	P_M	P_D	E[ΔX⁴] / (σ⁴Δt²)
1	0.02000	0.50250	0.00000	0.49750	1
sqrt 2	0.02828	0.25177	0.50000	0.24823	2
sqrt 3	0.03464	0.16811	0.66667	0.16522	3
2	0.04000	0.12625	0.75000	0.12375	4

$\lambda = 1$ **is a binomial tree.** Then $p_m = 0$: the middle branch vanishes, and the coefficient is 1 instead of 3. That is why a binomial step is a poor copy of a normal step in its tails, even though the many steps of a binomial tree still converge.

Can a probability go negative? $p_d \geq 0$ needs $\dfrac{1}{\lambda^2} \geq \dfrac{\nu \Delta t}{\Delta x} = \dfrac{\nu\sqrt{\Delta t}}{\lambda\sigma}$, which comes to

$$\Delta t \leq \frac{\sigma^2}{\lambda^2 \nu^2}.$$

How large the step can be with λ = √3 before p_d turns negative.

RATE R, %	ν	Largest Δt, years
3	0.01	133.33
10	0.08	2.08
25	0.23	0.25

For ordinary rates this never binds. It is a reminder that the tree is a real set of probabilities only for steps small enough.

Pricing on the tree

The value at each node is the discounted average of its three children:

$$V_j^i = e^{-r\Delta t}\left(p_u V_{j+1}^{i+1} + p_m V_j^{i+1} + p_d V_{j-1}^{i+1}\right).$$

A small tree by hand. Two steps, $\Delta t = 0.5$, so $\Delta x = 0.2449$, $p_u = 0.1769$, $p_m = 0.6667$, $p_d = 0.1565$, discount 0.9851.

- At T: payoffs are 58.21 and 22.76 at the top two nodes, and 0 elsewhere.
- At the middle time: the node at 127.76 is worth $e^{-r\Delta t}(p_u \times 58.21 + p_m \times 22.76) = 25.0879$; the node at 100 is worth $e^{-r\Delta t} p_u \times 22.76 = 3.9649$; the bottom node is worth 0.
- At the start: $e^{-r\Delta t}(p_u \times 25.0879 + p_m \times 3.9649) = 6.9752$.

Not bad for two steps, against 7.1281. With more steps:

The trinomial tree with λ = √3. The errors of call and put are nearly equal, as put-call parity says they must be.

STEPS N	CALL	ERROR	PUT	ERROR	NODES
10	7.2233	0.0952	9.1205	0.0957	121
50	7.0887	-0.0394	8.9856	-0.0393	2,601
100	7.1369	0.0088	9.0337	0.0089	10,201
500	7.1271	-0.0010	9.0239	-0.0010	251,001
1,000	7.1290	0.0010	9.0258	0.0010	1,002,001

Against the binomial tree

A fair comparison. A trinomial tree of n steps has $(n+1)^2$ nodes. A binomial one has $(n+1)(n+2)/2$, about half. So give the binomial tree $\sqrt{2}$ times the steps. Judge by the worst error over ten neighbouring sizes.

The same number of nodes, roughly. Neither method wins clearly: the strike's position among the nodes matters more than the third branch.

TRINOMIAL STEPS	WORST ERROR	BINOMIAL STEPS	WORST ERROR
50	0.03935	71	0.02603
100	0.00900	141	0.01314
200	0.00984	283	0.00641
400	0.00114	566	0.00307
800	0.00247	1,131	0.00149

This may be a disappointment. The third branch alone isn't a large win for this option. The strike still falls in a different place among the final nodes each time.

Where the win is: use the freedom in λ. Since λ is free, we can make the strike sit **exactly** on a node: choose Δx so that $j\,\Delta x = \ln(K/S_0)$ for a whole number j. To keep λ the same as n grows, take the sizes $n = 50j^2$, which gives $\lambda = 1.725$ every time, very close to $\sqrt{3}$. Now the kink is always in the same place relative to the grid.

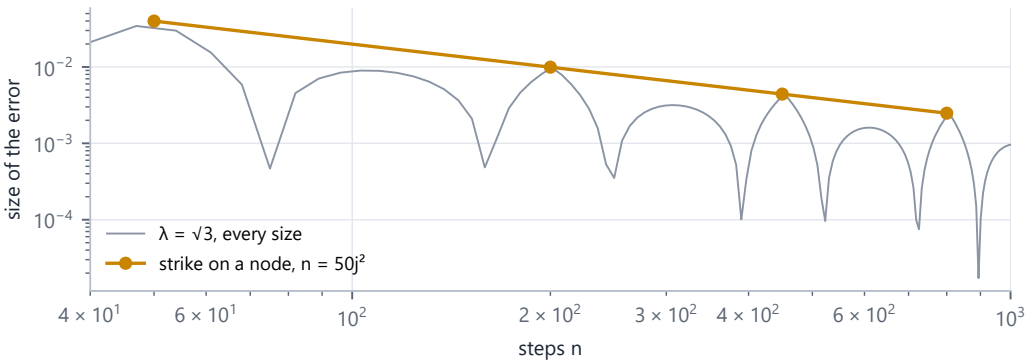


Figure 23.2. The size of the error against the number of steps. The grey curve uses λ = √3 at every size, and swings as the strike moves among the nodes. (It dips where the error changes sign.) Amber uses only the sizes with the strike on a node: the error is steady, falling exactly like 1/n.

Strike on a node, with λ = 1.725. Error × n hardly changes. The last column is $(4V_{4n} - V_n)/3$ minus the formula.

J	STEPS N	ERROR	ERROR × N	ERROR AT 4N	EXTRAPOLATED
1	50	-0.03987	-1.994	-0.00993	4.5e-05
2	200	-0.00993	-1.987	-0.00248	2.8e-06
3	450	-0.00441	-1.986	-0.00110	5.5e-07
4	800	-0.00248	-1.985	-0.00062	1.7e-07

Reading the table. The error at these sizes isn't smaller than a typical trinomial or binomial error. What has changed is that it's **predictable**: error × n stays at -1.99. And what's predictable can be removed. Since the steps go up by 4, the Richardson recipe of Lesson 22 is:

$$V \approx V_{4n} + \frac{V_{4n} - V_n}{3} = \frac{4V_{4n} - V_n}{3}.$$

The extrapolated values are hundreds to thousands of times more accurate than the raw ones: 2.8e-06 from trees of 200 and 800 steps. That is the real gain from the extra free parameter.

What to remember

- Work in $X = \ln S$: constant mean $\nu\Delta t$ and variance $\sigma^2\Delta t$ per step.
- $\Delta x = \lambda\sigma\sqrt{\Delta t}$; $p_m = 1 - 1/\lambda^2$; $p_{u,d} = \frac{1}{2}(1/\lambda^2 \pm \nu\Delta t/\Delta x)$.
- $\lambda = \sqrt{3}$ also matches the third and fourth moments; $\lambda = 1$ is a binomial tree.
- $p_d \geq 0$ needs $\Delta t \leq \sigma^2/(\lambda^2\nu^2)$.

- The third branch alone doesn't beat a binomial tree of equal size. Its freedom lets the strike sit on a node, which makes the error steady and removable by extrapolation.

Common mistakes

- **Comparing trees at equal steps, not equal work.** A trinomial tree has about twice the nodes.
- **Using $\sigma\sqrt{\Delta t}$ as the jump and calling the result trinomial.** Then $p_m = 0$.
- **Thinking matching more moments means more accuracy for every option.** The kink in the payoff often matters more.
- **Extrapolating from jumpy sizes.** Richardson needs a steady error.
- **Ignoring a negative probability.** The tree is then no longer an average, and values can blow up (Lesson 29).

Exercises

Solutions are in *Part 4: Solutions*.

23.1 Solve the three moment equations for p_u, p_m and p_d with $\Delta x = \lambda\sigma\sqrt{\Delta t}$, step by step. Evaluate them with $\lambda = \sqrt{3}$ and 100 steps.

23.2 Compute the third moment $E[\Delta X^3]$ on the tree exactly. A normal step with mean $m = \nu\Delta t$ and variance $s^2 = \sigma^2\Delta t$ has $E[\Delta X^3] = m^3 + 3ms^2$. For which λ do the two agree to leading order in Δt?

23.3 Show that the tree's fourth moment is $\lambda^2\sigma^4\Delta t^2$ to leading order, and that the normal's is $3\sigma^4\Delta t^2$. For which λ do they agree? Why does a binomial tree ($\lambda = 1$) get it wrong by a factor of 3?

23.4 Find how small Δt must be for $p_d \geq 0$ when $\lambda = \sqrt{3}$. What is the condition with the course's numbers? With $r = 25\%$? What goes wrong if a probability is negative?

23.5 Price the call by hand on a one-step trinomial tree ($\Delta t = 1$, $\lambda = \sqrt{3}$), using $S_0 = 100$, and compare with 7.13. Why is one step so poor? Check also that the average discounted share on this tree is close to 100.

23.6 Count the nodes in a trinomial tree of n steps and in a binomial tree of n steps. If one node's work is the same, how many binomial steps cost the same as n trinomial steps? What is the total memory if you keep only one column?

23.7 *(Code)* Build the trinomial pricer with $\lambda = \sqrt{3}$ and price the call and the put at $n = 10, 50, 100, 500$. Check put-call parity on the tree at each n: is it exact? Why, or why not?

23.8 *(Code)* Compare $\lambda = 1.2, \sqrt{2}, \sqrt{3}, 2$ for the call by the worst error over $n = 200, \dots, 209$. Then price the American put at $n = 100, 200, 400, 800$ on the $\lambda = \sqrt{3}$ tree and compare with Lesson 22's value.

Self-test

1. What are the three probabilities with $\lambda = \sqrt{3}$, before the small tilt?
2. Which λ turns the trinomial tree back into a binomial one?
3. Putting the strike on a node doesn't make the error smaller. Why is it useful anyway?

Companion notebook: `s5e23_trinomial_tree.ipynb`. Every number in this chapter is computed by it.

LESSON 24 · VIDEO 24

Delta Hedging in Practice

The hedging recipe, followed along one path; the error on each step, from Taylor and the Black-Scholes equation; why the final error's spread falls like $1/\sqrt{n}$; why the expected return makes little difference; and what happens when you price and hedge with the wrong volatility.

Why this lesson

Lesson 19 said: hold $\Delta = V_S$ shares at every instant, and the option is perfectly copied. Nobody can trade at every instant. So what actually happens when you rebalance once a day? Or once a month?

Simulating this is the most convincing test of the whole theory. **It works.** A seller who hedges ends with almost exactly what they owe, on nearly every path, whichever way the share went. It also shows clearly what's left over. The error has a definite size, set by the number of rebalances, and a definite cause: the difference between the realised and the assumed volatility.

What you need from earlier lessons

- **The hedge and the equation** (Lesson 19). $\Delta = V_S$; $\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rV$.
- **The Greeks** (Lesson 21). $\Delta = N(d_+)$, gamma, vega.
- **Quadratic variation** (Lesson 14). $\sum(\Delta \ln S)^2 \to \sigma^2 T$, and this is why realised volatility can be measured.

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$, $\mu = 8\%$. Sell a call for 7.1281.

The picture

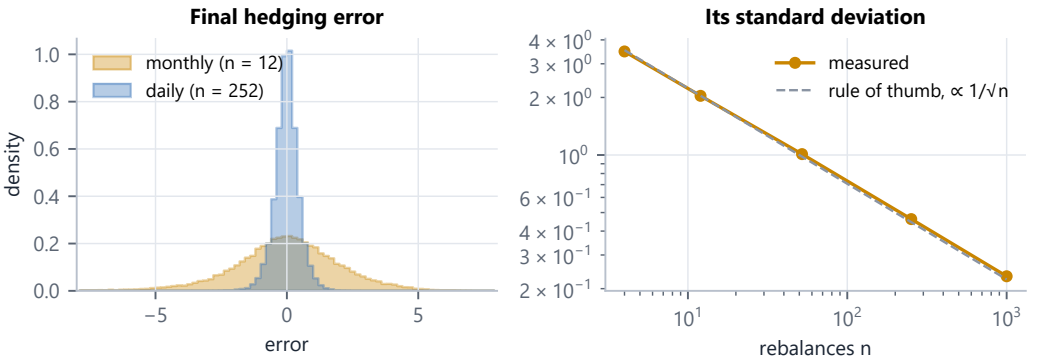


Figure 24.1. Left: what's left at expiry after selling the call and hedging, over 40,000 paths. A monthly hedge leaves a wide spread; a daily one, a narrow spike around zero. Right: the standard deviation against the number of rebalances, falling like 1/√n.

Notice what is *not* in the picture. These results include share paths that went far up and far down, with a real-world drift, and the hedger hardly notices.

The recipe, on one path

Start. Receive the option price. Buy $\Delta = N(d_+)$ shares. What's missing is borrowed.

At each rebalance. The cash grows by interest. Compute the new delta from the current price and time left. Trade the difference, paying for it from cash.

At expiry. Sell the shares, settle the loan, pay the option holder. What's left is the **hedging error**.

Twelve monthly steps on one simulated path:

One path, rebalanced monthly. The cash is negative: this hedge is financed with a loan.

TIME	SHARE	DELTA	CASH	OPTION VALUE
0.000	100.00	0.502	-43.11	7.13
0.083	96.00	0.410	-34.31	4.88
0.167	97.83	0.437	-37.06	5.22
0.250	88.13	0.213	-17.46	1.71
0.333	96.00	0.365	-32.08	3.59
0.417	100.11	0.452	-40.81	4.77
0.500	98.92	0.403	-36.14	3.73
0.583	97.65	0.344	-30.45	2.71
0.667	99.87	0.386	-34.72	2.93
0.750	98.83	0.316	-27.83	1.91
0.833	98.04	0.230	-19.55	1.03
0.917	102.72	0.379	-34.84	1.51

Reading it. The share falls to 88 by month 3, and delta falls with it, so the hedger sells into the fall. As the share climbs back, delta climbs too, so the hedger buys back at higher prices. **Selling low and buying high, every time: that is what hedging a sold option looks like.** This is the cost of the hedge, and the option price paid for it.

At expiry. The share ends at 106.34, so the option pays 1.34. The hedge holds shares worth 40.28 and cash of -34.93. The final error is 4.01. That's a fairly large one, about two standard deviations for a monthly hedge. A dozen rebalances aren't many.

The hedging error

Where does the error come from? Between rebalances the hedge is frozen, and the option's value curves away from it.

Follow the seller over one step, from S to $S + \delta S$, with the option's value by Taylor's theorem in two variables (Lesson 2).

Step 1. The option's change.

$$\delta V \approx \Theta\,\delta t + \Delta\,\delta S + \tfrac{1}{2}\Gamma(\delta S)^2.$$

Step 2. The hedge's change. Δ shares gain $\Delta\,\delta S$. The cash $(V - \Delta S)$ gains $r(V - \Delta S)\,\delta t$.

Step 3. Portfolio minus option. The seller's position changes by hedge minus option. The $\Delta\,\delta S$ cancels (that's what the hedge is for), leaving

$$r(V - \Delta S)\,\delta t - \Theta\,\delta t - \tfrac{1}{2}\Gamma(\delta S)^2.$$

Step 4. Use the equation. From Lesson 19, $rV - rS\Delta - \Theta = \tfrac{1}{2}\sigma^2 S^2\Gamma$. Substituting:

$$\boxed{\text{error on the step} \approx \tfrac{1}{2}\Gamma S^2\left(\sigma^2\delta t - \left(\tfrac{\delta S}{S}\right)^2\right)}$$

This is the key formula of the lesson. The seller gains when the squared move is smaller than the $\sigma^2\delta t$ that was priced in, and loses when it's larger. The sign of the move doesn't matter, only its size. Gamma sets the stakes.

Checked, over one week at $S = 100$:

The seller's error over one week, one year from expiry. It is an even function of the move, up to a small third-order correction.

MOVE, %	EXACT ERROR	FORMULA
0.00	0.0771	0.0767
1.00	0.0689	0.0667
-1.00	0.0651	0.0667
2.77	0.0059	0.0002
-2.77	-0.0061	0.0002
5.00	-0.1598	-0.1726
-5.00	-0.1872	-0.1726

The break-even move is $\sigma\sqrt{\delta t} = 2.77\%$ for a week. A share at 100 that moves by less than that in a week pays the seller; one that moves by more costs them.

Rebalancing more often

The errors on successive steps have mean close to zero, since $E[(\delta S/S)^2] \approx \sigma^2\delta t$. They are also nearly independent. So they add like a random walk (Lesson 13).

The scaling. Per step, the bracket $\sigma^2\delta t - (\delta S/S)^2$ has a standard deviation of about $\sqrt{2}\sigma^2\delta t$, because the variance of Z^2 is 2. So one step's error has standard deviation proportional to δt. Adding n independent steps, with $\delta t = T/n$, gives

$$\text{sd} \propto \sqrt{n} \times \frac{T}{n} = \frac{T}{\sqrt{n}}.$$

Four times the trades, half the error.

40,000 paths at each n. The mean is near zero, the sd halves as n goes up by 4.

REBALANCES N	MEAN ERROR	SD OF ERROR	SD, % OF PRICE	5TH PERCENTILE	Sd × √n
4	-0.0589	3.4797	48.8	-6.05	6.96
12	-0.0156	2.0391	28.6	-3.42	7.06
52	-0.0101	1.0107	14.2	-1.68	7.29
252	-0.0037	0.4622	6.5	-0.77	7.34
1,000	-0.0009	0.2322	3.3	-0.38	7.34

Reading the table. Hedging daily, the standard deviation is 0.46, or 6.5% of the price. From $n = 252$ to $n = 1000$ the ratio is 1.99: almost exactly 2, as predicted. The product sd × $\sqrt{n}$ settles to about 7.3. A rule of thumb for near-the-money options is $\sqrt{\pi/4}\,\sigma\mathcal{V} = 7.07$, which is close.

The limit. As $n \to \infty$, the error goes to 0. That's the exact replication of Lesson 19. In practice, trading costs stop you from going far. The choice of n is a trade-off: more trades, less risk, more cost.

The drift doesn't matter

Run the same hedge with different real-world expected returns, and for comparison, the seller who doesn't hedge at all, just holding the premium in cash:

Daily hedging, 20,000 paths at each μ, against the unhedged seller.

μ, %	HEDGED, MEAN	HEDGED, SD	UNHEDGED, MEAN	UNHEDGED, SD
-10	-0.0086	0.411	4.76	7.40
0	-0.0024	0.453	1.44	11.56
8	-0.0026	0.460	-2.90	15.37
20	-0.0143	0.444	-12.69	21.18
40	-0.0424	0.354	-37.16	29.53

Unhedged, μ is everything: the seller's average result ranges from 4.76 to -37.16. **Hedged,** the average stays near zero. The drift is simply not in the error formula above.

A fine point. The hedged spread isn't quite constant: it's 0.460 at $\mu = 8\%$ and 0.354 at $\mu = 40\%$. A strong drift carries the share away from the strike, where gamma is lower, so there's less to go wrong. This effect is second-order and doesn't affect the average:

there is no way to profit from it without a view on the direction of the share, and then you wouldn't be hedging.

The wrong volatility

So μ doesn't matter. Volatility does. What if you sell and hedge at 20%, while the share really moves at another volatility?

The error formula tells you. Summing over steps, the expected value of $(\delta S/S)^2$ is $\sigma_{\text{true}}^2 \delta t$, so the expected profit is

$$\tfrac{1}{2}(\sigma^2 - \sigma_{\text{true}}^2) \int_0^T (\text{discount})\, E[\Gamma_t S_t^2]\, dt.$$

(Exercise 24.4.) If the share is more volatile than you assumed, you lose. If less, you gain. Since Lesson 21 showed that vega is $\sigma\tau S^2\Gamma$, this is, to first order, simply the difference between the two prices.

Sold and hedged at 20%, rebalanced daily, 20,000 paths. The mean profit is close to the difference of the two Black-Scholes prices, carried to expiry.

TRUE VOL, %	MEAN PROFIT	BS DIFFERENCE, AT T	VEGA ESTIMATE, AT T	SD
15	2.085	2.055	2.055	0.752
18	0.834	0.822	0.822	0.510
20	-0.001	0.000	0.000	0.460
22	-0.836	-0.822	-0.822	0.603
25	-2.087	-2.054	-2.055	1.049
30	-4.166	-4.105	-4.111	2.029

Reading the table. Sell at 20% when the truth is 25%, and your average result is -2.09: you undercharged for the moves you'd have to hedge. Notice the last column too: the spread gets worse when the true volatility moves away from 20%, because the hedge itself uses the wrong delta.

This is why options trade on volatility. A hedged option is a position not on the direction of the share, but on how much it will move, compared with the volatility you paid for. The market's **implied volatility** is the σ that makes the formula match the market price; a trader who hedges is betting on realised against implied volatility.

What to remember

- **The recipe:** sell, buy Δ shares, rebalance, settle. Sell low and buy high: that's the cost the premium pays for.
- **One step:** error $\approx \frac{1}{2}\Gamma S^2(\sigma^2\delta t - (\delta S/S)^2)$. Sign of the move doesn't matter.
- **Break-even move:** $\sigma\sqrt{\delta t}$.
- **Spread:** sd of the final error $\propto 1/\sqrt{n}$; about 0.46 when hedging daily here.
- **μ:** no effect on the average.
- **Wrong volatility:** mean profit $\approx$ difference of the two BS prices; more realised volatility than priced means a loss.

Common mistakes

- **Expecting zero error.** It only shrinks with n. Discrete hedging always has some.
- **Thinking of hedging as "protection against a fall".** The hedge protects against moves of both signs and trades the direction risk for gamma risk.
- **Forgetting interest on the loan.** The cash account is usually a big negative number; ignoring its interest biases the result.
- **Using a stale delta.** The delta must use the current share price and time left.
- **Thinking the hedge removes all risk.** It removes direction risk. Volatility risk remains, and it's the main risk of a hedged option.

Exercises

Solutions are in *Part 4: Solutions*.

24.1 Using Taylor's theorem to second order and the Black-Scholes equation, derive the one-step error of a short-option hedger: $\frac{1}{2}\Gamma S^2\big(\sigma^2\Delta t - (\Delta S/S)^2\big)$.

24.2 Treat the one-step errors as independent, with ΓS^2 roughly constant. Using $\mathrm{Var}\big((\Delta S/S)^2\big) \approx 2\sigma^4\Delta t^2$, show that the standard deviation of the total error is proportional to $1/\sqrt{n}$. What should the ratio of the errors at $n = 252$ and $n = 1000$ be?

24.3 With one week between rebalances, how large a move leaves the hedger exactly even? What about one day? Compute both with $\sigma = 20\%$, and explain why the answer has nothing to do with the option.

24.4 The share really moves with volatility σ_{true}, and you sell and hedge at σ. Use the one-step error to find the expected profit per unit of time, and show that its present value is $\frac{1}{2}(\sigma^2 - \sigma_{\text{true}}^2)\int_0^T e^{-rt}E[S_t^2\Gamma_t]\,dt$. Which way does the profit go?

24.5 Explain from the one-step error why the expected return μ doesn't affect the hedger at leading order. Where does it enter at all? Why doesn't this help you pick shares?

24.6 Use vega to approximate the mean profit from selling at $\sigma = 20\%$ when the true volatility is 25%: $\text{mean} \approx \mathcal{V} \cdot (\sigma - \sigma_{\text{true}})$. With $\mathcal{V} = 39.89$, what is it, and how does it compare with the exact difference of the two Black-Scholes prices?

24.7 *(Code)* Hedge daily and record for each path the realised volatility $\sqrt{\sum (\Delta \ln S)^2 / T}$ and the final error. Sort the paths into buckets by realised volatility. Is the error related? What is the correlation, and does the relationship change if the path's final price is near the strike or far from it?

24.8 *(Code)* Delta-hedge a sold put (the delta is $N(d_+) - 1$) with $n = 12, 52, 252$. Show that the standard deviations are close to the call's, and explain why, using put-call parity.

Self-test

1. You quadruple the number of rebalances. What happens to the standard deviation of the hedging error?
2. What is the break-even daily move with $\sigma = 20\%$?
3. You sold at 20% volatility, and the share turns out to move at 25%. Do you make or lose money on average?

Companion notebook: `s5e24_delta_hedging.ipynb`. Every number in this chapter is computed by it.

LESSON 25 · VIDEO 25

Girsanov's Theorem

Reweighting paths, first on a coin; the density process Z and why its average is 1; a new probability built from the weights; the theorem, that W + θt becomes a Brownian motion, with a full proof; and why the change of probability can move the drift but never the volatility.

Why this lesson

Lesson 20 priced the call by letting the share drift at r, when in reality it drifts at μ. Why is that allowed? Part 2 gave the answer on a tree: change the probabilities of the paths. Here we do the same in continuous time, and the result is **Girsanov's theorem**. It tells you exactly how a change of probability affects a Brownian motion.

The headline: the new probability adds a constant drift to W and changes nothing else. That is precisely what's needed to turn μ into r.

Girsanov has a reputation for being hard. In fact the proof takes a few lines of the normal moment generating function. What's hard is the idea of changing probabilities without changing the paths, so we build that idea first on a coin.

What you need from earlier lessons

- **The Radon-Nikodym derivative** (Lesson 12). On a tree: $Z = \tilde{P}(\text{path})/P(\text{path})$, and $\tilde{E}[X] = E[ZX]$.
- **The moment generating function** (Lesson 4). $E[e^{uX}] = e^{mu+\frac{1}{2}s^2u^2}$ for $X \sim N(m, s^2)$.
- **The exponential martingale** (Lesson 18). $d\left(e^{\theta W_t - \frac{1}{2}\theta^2 t}\right) = \theta e^{\theta W_t - \frac{1}{2}\theta^2 t}\, dW_t$.
- **Quadratic variation** (Lesson 14). $\sum(\Delta W)^2 \to T$, on almost every path.

Numbers. $\theta = (\mu - r)/\sigma = (8\% - 3\%)/20\% = 0.25$, the share's market price of risk.

The picture

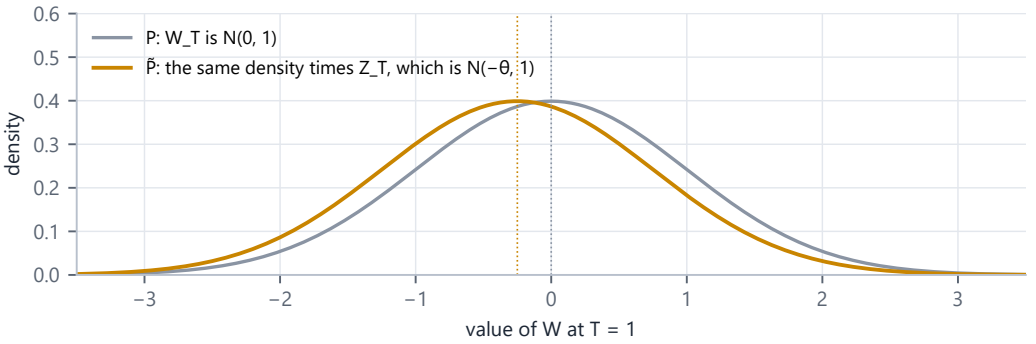


Figure 25.1. The density of W at time 1 under the original probability P (grey), and after multiplying by the weight Z_T (amber). The result is exactly another normal density, with mean moved to −θ and the same width.

Don't move the paths. Weigh them differently: paths that end low get more weight, and paths that end high get less. The bell curve shifts left without changing shape. This is the same calculation as the completed square of Lesson 20.

Reweighting, on a coin first

Toss a coin twice. Under P, heads has probability 0.6. Under a new probability $\tilde{P}$, heads has probability 0.4. The paths are the same; only their probabilities differ. The weight of a path is the ratio $Z = \tilde{P}/P$:

Two tosses. The weight Z is large where the new probability exceeds the old.

PATH	HEADS	P	P̃	Z = P̃/P
HH	2	0.36	0.16	0.4444
HT	1	0.24	0.24	1.0000
TH	1	0.24	0.24	1.0000
TT	0	0.16	0.36	2.2500

Two things to notice:

- **The weights average 1 under P:** $E[Z] = 1.0$, since the probabilities $\tilde{P}$ add up to 1.
- **Any average under $\tilde{P}$ is a weighted average under P.** For the number of heads, $\tilde{E}[\text{heads}] = 0.8$ directly, and $E[Z \times \text{heads}] = 0.8$.

This is what will carry over. With Brownian motion, each path has zero probability, so we can't divide probabilities path by path, but we can still write down the weight.

The density process Z

Definition. For a constant θ, let

$$Z_t = \exp\Big(-\theta W_t - \tfrac{1}{2}\theta^2 t\Big).$$

This is the exponential martingale of Lesson 18, with $-\theta$ in place of θ.

It is positive, and $Z_0 = 1$.

It has average 1. $W_t \sim N(0,t)$, so by the moment generating function, $E[e^{-\theta W_t}] = e^{\frac{1}{2}\theta^2 t}$. Hence $E[Z_t] = e^{\frac{1}{2}\theta^2 t} e^{-\frac{1}{2}\theta^2 t} = 1$.

It is a martingale. Itô's lemma (Exercise 25.2) gives

$$dZ_t = -\theta Z_t \, dW_t.$$

There's no dt term, so the average stays put (Lesson 18).

In numbers. With $\theta = 0.25$ at $T = 1$: a path ending at $W_T = 1$ has weight 0.7548, one ending at $W_T = -1$ has 1.2445, and one ending at 0 has 0.9692. Over 200,000 simulated paths, the average of Z_t at $t = 0.2, 0.5, 1$ is 1.0003, 1.0006 and 1.0003. The variance of Z_T is 0.0647 (exactly, $e^{\theta^2 T} - 1 = 0.0645$).

A new probability from the weights

Definition. For any event A that depends on the path up to T,

$$\tilde{P}(A) = E[Z_T \mathbf{1}_A].$$

And more generally, $\tilde{E}[X] = E[Z_T X]$. Exercise 25.3 checks that this is a probability.

Same possible events. Since $Z_T > 0$ always, an event has probability zero under P exactly when it has probability zero under $\tilde{P}$. The two probabilities are called **equivalent**. They agree on what can happen and disagree only on how likely it is. A **Radon-Nikodym theorem** says that any probability equivalent to P arises from some positive Z with average 1; here we need only this explicit one.

What the weights do to W_T. Multiply the density of W_T by the weight, and complete the square:

$$\begin{aligned} \frac{e^{-w^2/(2T)}}{\sqrt{2\pi T}} \, e^{-\theta w - \frac{1}{2}\theta^2 T} &= \frac{1}{\sqrt{2\pi T}} e^{-(w^2 + 2\theta T w + \theta^2 T^2)/(2T)} \\ &= \frac{1}{\sqrt{2\pi T}} e^{-(w+\theta T)^2/(2T)}. \end{aligned}$$

Under $\tilde{P}$, W_T is normal with mean $-\theta T$ and variance T, the curve in the picture. Weighted simulated paths give a mean of -0.2514 (exact -0.25) and a variance of 1.0024.

The same simulated paths, with and without the weights. The events don't change. Their probabilities do.

EVENT	P, SIMULATED	P, EXACT	P̃, WEIGHTED	P̃, EXACT
W_T > 0	0.5008	0.5000	0.4019	0.4013
W_T > 1	0.1589	0.1587	0.1059	0.1056
W_T < -1	0.1598	0.1587	0.2283	0.2266

Girsanov's theorem

So W_T has mean $-\theta T$ under $\tilde{P}$. Add the missing drift back, and you'd expect something with mean zero. The theorem says much more than that.

> **Girsanov's theorem.** Under $\tilde{P}$, the process $\tilde{W}_t = W_t + \theta t$ is a standard Brownian motion.

Proof. A Brownian motion is defined by continuous paths starting at 0 with independent, normal increments of variance equal to the elapsed time (Lesson 13). $\tilde{W}$ is continuous and starts at 0. So it's enough to check the increments, and we use the moment generating function. Take $s < t$.

Step 1. Write the weight in two pieces. Let $\mathcal{F}_s$ stand for the path up to s. Since Z is a martingale, an average under $\tilde{P}$ conditional on $\mathcal{F}_s$ can be written with the ratio of the weights:

$$\tilde{E}\left[e^{u(\tilde{W}_t - \tilde{W}_s)} \mid \mathcal{F}_s\right] = E\left[\frac{Z_t}{Z_s} e^{u(\tilde{W}_t - \tilde{W}_s)} \,\middle|\, \mathcal{F}_s\right].$$

Step 2. Substitute. With $D = W_t - W_s$ and $h = t - s$: $\dfrac{Z_t}{Z_s} = e^{-\theta D - \frac{1}{2}\theta^2 h}$, and $\tilde{W}_t - \tilde{W}_s = D + \theta h$.

Step 3. Combine exponents. Inside the average: $e^{(u-\theta)D + u\theta h - \frac{1}{2}\theta^2 h}$.

Step 4. Average. D is independent of $\mathcal{F}_s$ and is $N(0, h)$, so $E[e^{(u-\theta)D}] = e^{\frac{1}{2}(u-\theta)^2 h}$.

Step 5. Add up. The total exponent is

$$\tfrac{1}{2}(u^2 - 2u\theta + \theta^2)h + u\theta h - \tfrac{1}{2}\theta^2 h = \tfrac{1}{2}u^2 h.$$

Everything cancels except $\frac{1}{2}u^2 h$. So

$$\tilde{E}\left[e^{u(\tilde{W}_t-\tilde{W}_s)} \mid \mathcal{F}_s\right] = e^{\frac{1}{2}u^2 h},$$

the moment generating function of $N(0,h)$, whatever $\mathcal{F}_s$. So the increment is $N(0,h)$ under $\tilde{P}$ and independent of the past. ■

Checked, on the weighted paths.

QUANTITY	WEIGHTED ESTIMATE	SHOULD BE
Mean of $\tilde{W}_{0.5}$	-0.0023	0
Variance of $\tilde{W}_{0.5}$	0.5008	0.5
Mean of $\tilde{W}_1$	-0.0014	0
Variance of $\tilde{W}_1$	1.0024	1
Covariance of the two half-year increments	-0.0003	0
$\tilde{E}[e^{0.7\tilde{W}_1}]$	1.2773	1.2776

Without the weights, the mean of $\tilde{W}_1$ would be 0.2492: not zero at all.

What it does to the share. Substitute $dW = d\tilde{W} - \theta\,dt$ in the share's equation:

$$dS = \mu S\,dt + \sigma S(d\tilde{W} - \theta\,dt) = (\mu - \sigma\theta)S\,dt + \sigma S\,d\tilde{W}.$$

With $\theta = (\mu - r)/\sigma$, we have $\mu - \sigma\theta = r$. So:

$$dS = rS\,dt + \sigma S\,d\tilde{W}.$$

This is the risk-neutral share, derived. The same paths, now counted with different weights, have drift r.

The same simulated shares, drifting at μ = 8% under P.

QUANTITY	UNDER P	UNDER P̃, WEIGHTED	P̃, EXACT
Average share price at T	108.3139	103.0500	103.0455
Probability that it ends above K	0.5222	0.4229	0.4231

The call's price, from the same real-world paths, weighted by Z_T, comes to 7.1396, with the formula at 7.1281. **The real-world paths price the option, once they're weighted.**

Volatility can't be changed

$\tilde{P}$ changed the drift of W. Could a clever choice of weights change the volatility too? No.

Recall (Lesson 14) that $\sum(\Delta W)^2 \to T$ as the steps shrink, **on almost every path**. Not on average, but on the paths themselves. The change of probability only reweights paths; it doesn't alter any path. So a statement about the paths that holds with probability 1 under P must hold with probability 1 under the equivalent $\tilde{P}$. The quadratic variation of W is T under both.

Checked. On 4,000 paths of 2,000 steps: the mean of $\sum(\Delta W)^2$ under P is 1.0006, and with the weights it's 1.0007. The correlation between the realised sum and the weight is 0.003. The sum is essentially a fixed number, and the weights can't move it.

Adding a drift doesn't change it either: $\sum(\Delta W + \theta\Delta t)^2$ differs from $\sum(\Delta W)^2$ by only 0.0002 on average, since the drift terms are of order Δt.

The consequence for finance. σ is measured from a path, by its realised variance. A change of probability can't touch it. That's why real-world and risk-neutral models share the same σ, and differ only in the drift. It's also why two equivalent probabilities can only disagree on μ.

A warning about large changes

Weights have a cost in simulation. The **effective sample size** of weighted paths is about $M/E[Z_T^2] = Me^{-\theta^2 T}$. For $\theta = 0.25$ that's 0.939 of the paths, which agrees with the theory, 0.939. A small change of probability costs almost nothing. A large one can leave a handful of paths with nearly all the weight (Exercises 25.6 and 25.7).

What to remember

- **The weight:** $Z_t = e^{-\theta W_t - \frac{1}{2}\theta^2 t}$; positive, starts at 1, martingale, $dZ = -\theta Z\, dW$.
- **The new probability:** $\tilde{P}(A) = E[Z_T \mathbf{1}_A]$; $\tilde{E}[X] = E[Z_T X]$. Equivalent to P.
- **Girsanov:** under $\tilde{P}$, $\tilde{W}_t = W_t + \theta t$ is a Brownian motion.
- **The share:** with $\theta = (\mu - r)/\sigma$, $dS = rS\, dt + \sigma S\, d\tilde{W}$.
- **Volatility is untouched:** quadratic variation holds path by path, so it's the same under both probabilities.
- The paths don't change. Only their weights do.

Common mistakes

- **Thinking the paths are moved.** The same paths, with different weights. It's W's *description* that changes: under $\tilde{P}$ it has a drift of $-\theta$.

- **Sign confusion.** With $Z_T = e^{-\theta W_T - \cdots}$, W has drift $-\theta$ under $\tilde{P}$, and $W + \theta t$ is the Brownian motion.
- **Trying to change volatility by Girsanov.** It can't be done with an equivalent probability.
- **Weighting by Z_t when pricing at T.** The weight for an event at T is Z_T.
- **Forgetting the $-\frac{1}{2}\theta^2 t$.** Without it, the weights don't average 1.

Exercises

Solutions are in *Part 4: Solutions*.

25.1 Show directly that $E[Z_T] = 1$ for $Z_T = e^{-\theta W_T - \frac{1}{2}\theta^2 T}$, using the moment generating function of the normal. Then show that Z_t is a martingale by splitting $W_T = W_t + (W_T - W_t)$.

25.2 Use Itô's lemma to show $dZ_t = -\theta Z_t \, dW_t$. Why does having no dt term make it a martingale? Find Z_T at $W_T = 1$ and $W_T = -1$ with $\theta = 0.25$.

25.3 Check that $\tilde{P}(A) = E[Z_T \mathbf{1}_A]$ is a probability: non-negative, total 1, additive on disjoint events. Why is it equivalent to P? What would break if Z_T could be 0?

25.4 Compute $\tilde{E}[e^{u\tilde{W}_T}]$ as $E[Z_T e^{u(W_T + \theta T)}]$ by combining exponents. Conclude that $\tilde{W}_T \sim N(0, T)$ under $\tilde{P}$.

25.5 Rewrite $dS = \mu S \, dt + \sigma S \, dW$ in terms of $d\tilde{W}$ with $\theta = (\mu - r)/\sigma$. Find S_T in terms of $\tilde{W}_T$, and compute $\tilde{P}(S_T > K)$ and $P(S_T > K)$ with the course's numbers.

25.6 Compute $E[Z_T^2] = e^{\theta^2 T}$. Importance-sampling theory says that weights with this second moment are worth about $e^{-\theta^2 T}$ of their paths in effective sample size. Evaluate this for $\theta = 0.25$ and for $\theta = 3$. Why is a large change of probability dangerous for a simulation?

25.7 *(Code)* Price the call and the put by simulating under P with $\mu = 8\%$ and weighting each path by Z_T; compare with simulating directly under $\tilde{P}$ and with the formula. Repeat with $\mu = 50\%$. Which standard errors shrink, which grow, and why?

25.8 *(Code)* On 2000 paths of 2000 steps, show that the realised $\sum (\Delta W)^2$ is unchanged by the change of probability, in mean and distribution, by computing its weighted mean and weighted quantiles. Then simulate $\tilde{W}$ and check that $\tilde{W}_t^2 - t$ has weighted mean 0 at $t = 0.5$ and $t = 1$.

Self-test

1. With $\theta = 0.25$, what is the mean of W_1 under $\tilde{P}$?
2. Which process is a Brownian motion under $\tilde{P}$?
3. Why can a change of probability not change the volatility?

Companion notebook: `s5e25_girsanov.ipynb`. Every number in this chapter is computed by it.

LESSON 26 · VIDEO 26

Risk-Neutral Pricing

Why the discounted share is a martingale under the new probability, and every self-financing portfolio with it; the pricing formula, proved; four ways to compute a price, one of them wrong; the martingale representation theorem and where the hedge comes from; and the two fundamental theorems of asset pricing.

Why this lesson

Part 2 proved risk-neutral pricing on a tree. Lesson 20 used it in continuous time, taking it on trust. This lesson proves it, and ties the threads together.

The logic is short:

1. Under the Girsanov probability $\tilde{P}$, the discounted share has no drift.
2. So every self-financing portfolio, discounted, has no drift either, whatever it holds.
3. If a portfolio copies an option, the option's price is the portfolio's value. So the discounted option has no drift.
4. A process with no drift is a martingale. Its value today is the average of its final value. **That's the pricing formula.**

Step 3 leaves one question: **does a copying portfolio always exist?** The martingale representation theorem says yes, and in doing so tells you the hedge. That's where we finish, with the two statements that sum up modern pricing theory.

What you need from earlier lessons

- **Girsanov** (Lesson 25). With $\theta = (\mu - r)/\sigma$ and $\tilde{W}_t = W_t + \theta t$, $\tilde{W}$ is Brownian under $\tilde{P}$.
- **The product rule** (Lesson 18). $d(XY) = X\,dY + Y\,dX + dX\,dY$.
- **The martingale test** (Lesson 18). No dt term means a martingale, given integrability.
- **The self-financing rule** (Lesson 19). $dX = \Delta\,dS + r(X - \Delta S)\,dt$.
- **The arbitrage arguments of Lesson 9.**

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$, $\mu = 8\%$, so $\theta = 0.25$.

The picture

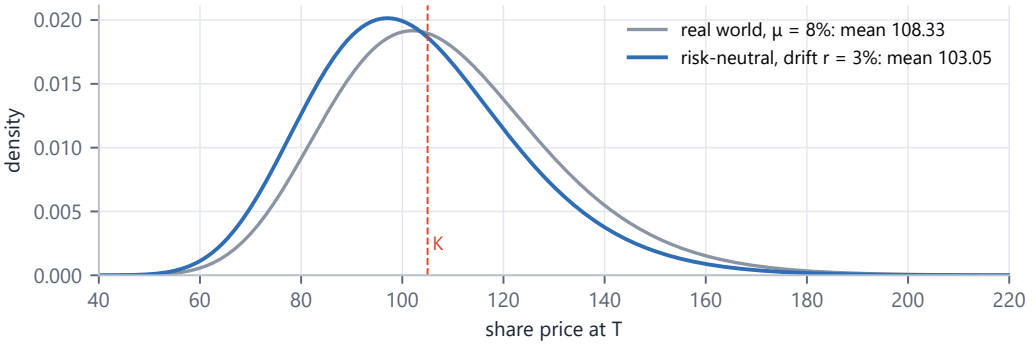


Figure 26.1. The same share at T, under two probabilities. Grey is the real world, with mean above 108. Blue is the risk-neutral world, with mean about 103. The price uses the blue curve, even though the world follows the grey one.

The blue curve isn't a forecast. It's a calculating device, a set of weights that gives the cost of the hedge.

The discounted share becomes a martingale

Let $D_t = e^{-rt}$, the **discount factor**. It has $dD = -rD\,dt$, with no randomness.

Step 1. The product rule. There is no cross-term, since dD has no dW:

$$d(D_tS_t) = D_t\,dS_t + S_t\,dD_t = D_t(\mu S_t\,dt + \sigma S_t\,dW_t) - rD_tS_t\,dt.$$

Step 2. Collect, and pull out σD_tS_t:

$$d(D_tS_t) = \sigma D_tS_t\left(\frac{\mu - r}{\sigma}\,dt + dW_t\right) = \sigma D_tS_t(\theta\,dt + dW_t).$$

So θ appears on its own. It is the extra return per unit of risk, the share's **Sharpe ratio**, also called its **market price of risk**: here $(8\% - 3\%)/20\% = 0.25$.

Step 3. Girsanov. $\theta\,dt + dW_t = d\tilde{W}_t$, so

$$\boxed{d(D_tS_t) = \sigma D_tS_t\,d\tilde{W}_t}$$

There's no dt term, so under $\tilde{P}$ the discounted share is a martingale.

Checked on 400,000 real-world paths:

The average discounted share. Under P it grows at μ − r = 5% a year. With the weights, it stays at 100.

TIME	UNDER P	EXACT	Under P̃ (weighted)	EXACT
0.25	101.257	101.258	99.978	100.000
0.50	102.526	102.532	99.975	100.000
0.75	103.819	103.821	99.979	100.000
1.00	105.144	105.127	99.996	100.000

Every self-financing portfolio too

From Lesson 19, a self-financing portfolio has $dX = \Delta\, dS + r(X - \Delta S)\, dt$. Apply the product rule to $D_t X_t$:

$$d(D_t X_t) = D_t\big(\Delta\, dS + r(X - \Delta S)\, dt\big) - rD_t X\, dt = \Delta\, D_t(dS - rS\, dt) = \Delta_t\, d(D_t S_t).$$

Your discounted wealth changes only through your position in the discounted share. Now use the boxed result:

$$d(D_t X_t) = \Delta_t \sigma D_t S_t\, d\tilde{W}_t.$$

No dt term, for any Δ_t. So $D_t X_t$ is a $\tilde{P}$-martingale. **No trading strategy changes the average discounted wealth under $\tilde{P}$.**

Four strategies, each starting from wealth 10:

Monthly rebalancing, one year. In the real world, holding shares earns more on average. Under P̃, every strategy averages 10.

STRATEGY	REAL-WORLD AVERAGE, DISCOUNTED	UNDER P̃
Cash only	10.000	9.998
Hold 1 share	15.144	10.012
Momentum: 2 shares above 100, none below	15.976	10.017
Contrarian: 2 shares below 100, none above	13.487	10.019

(The last column carries a little simulation noise.) Real-world, strategies do differ, as they should: risk is rewarded. Under $\tilde{P}$, nobody beats anybody.

Why that rules out free money. Suppose a strategy started from 0, never lost, and sometimes gained. Then its discounted wealth would average above 0 under $\tilde{P}$. But it's a martingale starting at 0, so its average must be 0. Contradiction. This is the "no arbi-

trage" half of the story.

(A technical point: Δ must not look into the future, and must not be so wild that the integral stops being a true martingale. Strategies such as doubling your bet forever are excluded by the integrability condition of Lesson 18.)

The pricing formula

Suppose a self-financing portfolio ends with $X_T = V_T$, the option's payoff. Then, by no arbitrage, the option's value at each earlier time is $V_t = X_t$. And $D_t X_t$ is a $\tilde{P}$-martingale, so

$$D_t V_t = \tilde{E}[D_T V_T \mid \mathcal{F}_t].$$

Divide by D_t:

$$\boxed{V_t = \tilde{E}\left[e^{-r(T-t)} V_T \,\middle|\, \mathcal{F}_t\right]}$$

This is the risk-neutral pricing formula. It applies to any payoff that depends on the path up to T, as long as it can be replicated.

Four ways to compute a price.

The call, with 2,000,000 paths. The first three agree. The last is a different quantity, and is not a price.

WAY	PRICE	STANDARD ERROR
The Black-Scholes formula	7.1281	0.0000
Simulate under the risk-neutral probability	7.1218	0.0089
Simulate under the real probability, weight by Z	7.1228	0.0066
Simulate under the real probability, no weights (wrong)	9.9515	0.0105

The last line discounts the real-world average payoff. It's 2.82 too high. That extra is a compensation for risk that a hedger doesn't bear, so nobody could charge it: a competitor would sell the option for less, hedge it, and lock in a profit.

A check on a forward. A contract paying $S_T - F$ at T is worth $\tilde{E}[e^{-rT}(S_T - F)] = S_0 - e^{-rT}F$, since the discounted share is a martingale. It's worth zero at $F = S_0 e^{rT}$, which is the no-arbitrage forward price. (Exercise 26.4.)

Where the hedge comes from

Now step 3's question: **does a copying portfolio always exist?** Given any payoff V_T at T, define

$$M_t = \tilde{E}[D_T V_T \mid \mathcal{F}_t].$$

This is a $\tilde{P}$-martingale, by the tower property (Lesson 8), and it ends at the discounted payoff.

> **The martingale representation theorem.** On a world driven by a single Brownian motion $\tilde{W}$, every such martingale can be written as $M_t = M_0 + \int_0^t \Gamma_u \, d\tilde{W}_u$ for some adapted process Γ.

(We don't prove this theorem; it's deep. It says that there's no other source of randomness for M to move with.)

Build the hedge. We want $D_t X_t = M_t$. We know $d(D_t X_t) = \Delta_t \sigma D_t S_t \, d\tilde{W}_t$, and $dM_t = \Gamma_t \, d\tilde{W}_t$. Matching:

$$\boxed{\Delta_t = \frac{\Gamma_t}{\sigma D_t S_t}}$$

Start with $X_0 = M_0$, hold this many shares, and the portfolio finishes at $X_T = V_T$. **So every payoff can be replicated.**

For the call. Itô on $D_t V(t, S_t)$ (Exercise 26.5 does a simpler case) gives $\Gamma_t = \sigma D_t S_t V_S$, and hence $\Delta_t = V_S = N(d_+)$. This agrees with Lesson 19's result, reached in a very different way.

Checked on one path. Rebuild M_t from its integral, with $n = 5000$ steps, along a risk-neutral path:

One path, finishing at about 119. Starting from the option's price and adding up the gains on the hedge reproduces the option's discounted value.

TIME	DISCOUNTED OPTION VALUE	$M_0 + \int \Gamma \, d\tilde{W}$
0.20	7.443	7.323
0.50	1.521	1.500
0.80	6.867	6.791
1.00	12.934	12.898

At the end, the discounted payoff is 13.356, and the rebuilt value is 13.320. The gap of -0.036 comes from using finitely many steps, and it's in line with Lesson 24.

The two fundamental theorems

Everything so far condenses into two statements. A probability $\tilde{P}$, equivalent to P, under which all discounted asset prices are martingales, is called a **risk-neutral probability**, or **equivalent martingale measure**.

> **First fundamental theorem of asset pricing.** If a risk-neutral probability exists, there is no arbitrage. (The converse also holds, under technical conditions.)

We proved the "if" part above. For a tree, Lesson 11 did so too.

> **Second fundamental theorem.** If there is no arbitrage, the market is **complete** (every payoff can be hedged) exactly when the risk-neutral probability is unique.

Here, one share and one Brownian motion give exactly one θ, so one $\tilde{P}$, and representation gives the hedge. If there were more sources of randomness than assets (a stochastic volatility, say), there would be many $\tilde{P}$'s, and many consistent prices. Some risk then can't be hedged away.

What if two assets disagree? Two assets driven by the same W must have the same θ, or there is free money. Take asset 1 with $(\mu, \sigma) = (0.08,\ 0.20)$ and asset 2 with $(0.14,\ 0.30)$. Their market prices of risk are 0.250 and 0.367.

Put weights 3 and -2 of your wealth in the two assets. The volatility is $3 \times 0.2 + (-2) \times 0.3 = 0$: no risk at all. Yet it earns -0.04, less than r. So do the reverse: short this riskless portfolio, invest in cash at 3%, and collect the difference of 7% a year, with no risk and no money down. No arbitrage forces $\mu_2 = r + \theta_1 \sigma_2 = 0.105$.

What to remember

- $d(D_t S_t) = \sigma D_t S_t (\theta\, dt + dW_t) = \sigma D_t S_t\, d\tilde{W}_t$: a $\tilde{P}$-martingale. θ is the Sharpe ratio.
- $d(D_t X_t) = \Delta_t\, d(D_t S_t)$: every self-financing portfolio, discounted, is a $\tilde{P}$-martingale.
- **Pricing:** $V_t = \tilde{E}[e^{-r(T-t)} V_T \mid \mathcal{F}_t]$.
- **Representation** gives Γ; the **hedge** is $\Delta_t = \Gamma_t / (\sigma D_t S_t)$; for a call, $\Delta = N(d_+)$.
- **First theorem:** a risk-neutral probability means no arbitrage. **Second:** unique means complete.
- Assets on the same Brownian motion must share one θ.

Common mistakes

- **"Investors are risk-neutral."** They aren't. $\tilde{P}$ is a calculating device; the real world is P.
- **Discounting a real-world average.** That gives 9.95 for the call, and it's not a price.
- **Thinking $\tilde{P}$ is a forecast.** Use P for forecasts. Use $\tilde{P}$ for prices.
- **Assuming every market is complete.** With more random sources than assets, the hedge isn't perfect and the price isn't unique.
- **Skipping the discount.** It's $D_t V_t$, not V_t, that's a martingale.

Exercises

Solutions are in *Part 4: Solutions*.

26.1 Use the product rule on $D_t S_t$ to derive $d(D_t S_t) = \sigma D_t S_t(\theta\, dt + dW_t)$, with $\theta = (\mu - r)/\sigma$. Why is there no cross-variation term between D and S?

26.2 θ is the share's **Sharpe ratio**, excess return per unit of risk. Evaluate it with the course's numbers. What does $\theta = 0$ mean, and what would $\tilde{P}$ be then? What is $E[S_T]$ under $\tilde{P}$ and under P?

26.3 Show that $d(D_t X_t) = \Delta_t\, d(D_t S_t)$ for a self-financing portfolio, and therefore that $D_t X_t$ is a $\tilde{P}$-martingale. Why does this rule out making money for sure, whatever Δ_t is?

26.4 Use the pricing formula to find the value today of a forward contract paying $S_T - F$ at T. Which F makes it worth zero? Evaluate it with the course's numbers. Did you need the volatility? What about μ?

26.5 Find the martingale representation of $\tilde{W}_T^2 - T$: use Itô's lemma to write it as $\int_0^T 2\tilde{W}_t\, d\tilde{W}_t$. If this were the discounted value of a claim, how many shares would the hedge hold at time t, with $r = 0$?

26.6 Two assets are driven by one Brownian motion: $(\mu_1, \sigma_1) = (8\%, 20\%)$ and $(\mu_2, \sigma_2) = (14\%, 30\%)$. Build the portfolio that removes the dW term. What does it earn? What arbitrage does that allow? Which μ_2 removes it?

26.7 *(Code)* Price the call and a cash digital on the same paths under P with $\mu = 8\%$, weighting by Z_T, and under $\tilde{P}$ directly. Compare standard errors for each with the formula. Which suffers more from the weights, and why?

26.8 *(Code)* Check the representation for the call on 200 paths of 2000 steps: compute $M_0 + \sum \Gamma\, \Delta\tilde{W}$ and compare with the discounted payoff path by path. Report the root-mean-square gap at 500, 1000 and 2000 steps. How fast does it shrink?

Self-test

1. What is θ here, and what does it measure?
2. Under $\tilde{P}$, what is the dt term of a self-financing portfolio's discounted value?
3. What does the martingale representation theorem give that the pricing formula alone doesn't?

Companion notebook: `s5e26_risk_neutral_pricing.ipynb`. Every number in this chapter is computed by it.

LESSON 27 · VIDEO 27

Change of Numéraire

Counting wealth in shares instead of dollars; the share-measure weight and how Girsanov moves the drift to r + σ²; prices in shares as martingales; what N(d₊) and N(d₋) really are, one probability for each unit; and Margrabe's exchange-option formula, a price with no interest rate in it.

Why this lesson

The Black-Scholes formula has two probabilities in it, $N(d_+)$ and $N(d_-)$. Lesson 20 showed that $N(d_-)$ is the risk-neutral chance of exercise, and that $N(d_+)$ came from completing a square. It looked like an accident of algebra. It isn't: $N(d_+)$ **is also a chance of exercise, under a different probability.**

That other probability arises from changing the unit you count money in. The idea is simple and very powerful. By choosing the right unit, you can make a hard problem easy. Here it gives a one-line derivation of the formula, and a price for an option to swap one asset for another. The same trick is how interest-rate options come to have Black-Scholes-style formulas.

What you need from earlier lessons

- **Risk-neutral pricing** (Lesson 26). $D_t S_t$ is a $\tilde{P}$-martingale; $V_0 = \tilde{E}[e^{-rT} V_T]$.
- **Girsanov** (Lesson 25). The weight $Z_T = e^{-\theta \tilde{W}_T - \frac{1}{2}\theta^2 T}$ makes $\tilde{W}_t + \theta t$ a Brownian motion.
- **The formula** (Lesson 20). $C = S_0 N(d_+) - K e^{-rT} N(d_-)$.
- **Correlated Brownian motions** (Lesson 14). $dW^1\, dW^2 = \rho\, dt$.

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$.

The picture

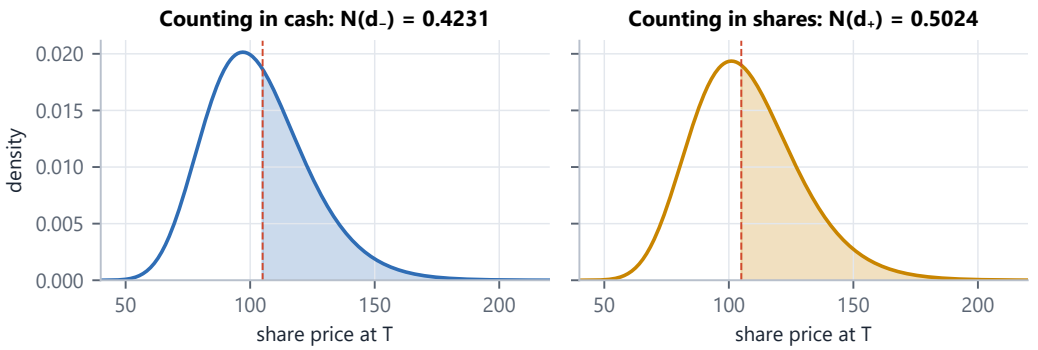


Figure 27.1. The share at T under two probabilities, each shaded above the strike. In cash units the shaded area is N(d₋). In share units, the curve moves right, because paths on which the share ends high count for more, and the shaded area becomes N(d₊).

Same paths, same event, same strike. The two areas differ only because the two units weigh the paths differently.

Counting in shares instead of dollars

A **numéraire** is the unit you count wealth in. So far the unit has been the cash account e^{rt}: Lesson 26's statement is that anything *divided by the cash account* is a $\tilde{P}$-martingale.

But you could count in shares. Measured in shares, wealth X is worth X/S. **In these units, the share is riskless:** one share is worth exactly 1 share, today and forever. Cash is what's risky, because its value in shares goes up and down.

One outcome in both units. Suppose the share ends at 120. The call pays 15 dollars, or $15/120 = 0.125$ shares. A dollar held in cash becomes 1.0305 dollars, or 0.00859 shares: fewer than the 0.01 shares it was worth at the start, since the share rose by more than the interest.

The share-measure weight

To count in shares we need the matching probability, under which things counted in shares are martingales. Define it with a weight on $\tilde{P}$:

$$Z_T^S = \frac{e^{-rT} S_T}{S_0}, \qquad P^S(A) = \tilde{E}[Z_T^S \mathbf{1}_A].$$

It's a valid weight. It is positive, and its average is 1, because the discounted share is a $\tilde{P}$-martingale: $\tilde{E}[e^{-rT}S_T] = S_0$. Simulated: mean 1.0000, variance 0.0408 (exact $e^{\sigma^2 T} - 1 = 0.0408$).

In words. Paths on which the share ends high count for more, in proportion to the share's discounted value.

It's a Girsanov weight. Under $\tilde{P}$, $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\tilde{W}_T}$. So

$$Z_T^S = e^{\sigma\tilde{W}_T - \frac{1}{2}\sigma^2 T}.$$

This is Lesson 25's weight with $\theta = -\sigma$. So Girsanov says:

> Under P^S, the process $W_t^S = \tilde{W}_t - \sigma t$ is a Brownian motion.

The share's motion under P^S. Put $d\tilde{W} = dW^S + \sigma\,dt$ in $dS = rS\,dt + \sigma S\,d\tilde{W}$:

$$dS = (r + \sigma^2)S\,dt + \sigma S\,dW^S.$$

The drift rises by σ^2. Volatility doesn't change, as Lesson 25 promised. With our numbers, the mean of $\ln(S_T/S_0)$ rises from $r - \frac{1}{2}\sigma^2 = 0.0100$ (simulated) to $r + \frac{1}{2}\sigma^2 = 0.05$ (weighted: 0.0500), and the median of S_T from 101.01 to 105.13.

Prices in shares are martingales

Claim. If V_t is the value of a traded payoff, then V_t/S_t is a P^S-martingale, so

$$\frac{V_0}{S_0} = E^S\left[\frac{V_T}{S_T}\right].$$

Proof. Use the definition of E^S, and let the weight cancel the denominator:

$$S_0 E^S\left[\frac{V_T}{S_T}\right] = S_0\tilde{E}\left[\frac{e^{-rT}S_T}{S_0}\cdot\frac{V_T}{S_T}\right] = \tilde{E}[e^{-rT}V_T] = V_0. \quad \blacksquare$$

Same answer, with no discounting. Pricing with the cash numéraire: discount by cash, average with $\tilde{P}$. Pricing with the share numéraire: divide by the share, average with P^S.

Checked: cash, counted in shares. e^{rt}/S_t should average $1/S_0 = 0.01$ under P^S:

Simulated directly under P^S, with drift r + σ².

TIME	Average of e^(rt)/S	$1/S_0$
0.25	0.01000	0.01000
0.50	0.01000	0.01000
1.00	0.01000	0.01000

(Exercise 27.3 proves it by Itô.)

Checked: the call. Simulating under P^S and averaging $S_0 \max(S_T - K, 0)/S_T$ gives 7.1227 (standard error 0.0070). Simulating under $\tilde{P}$ gives 7.1285 (standard error 0.0089). The formula gives 7.1281.

What N(d+) really is

Split the call into its two payoffs, as in Lesson 20, and price each with the numéraire that suits it.

The strike term: the payoff $K\mathbf{1}\{S_T > K\}$ is a fixed sum in cash, so use cash:

$$\tilde{E}[e^{-rT} K \mathbf{1}\{S_T > K\}] = K e^{-rT} \tilde{P}(S_T > K).$$

The share term: the payoff $S_T \mathbf{1}\{S_T > K\}$ is a fixed number of shares (one, or none), so use shares. By the claim above:

$$S_0 E^S\Big[\frac{S_T \mathbf{1}\{S_T > K\}}{S_T}\Big] = S_0 P^S(S_T > K).$$

So, with no integrals at all:

$$\boxed{C = S_0\, P^S(S_T > K) - K e^{-rT}\, \tilde{P}(S_T > K)}$$

Each probability, computed. Under P^S, $\ln S_T$ is normal with mean $\ln S_0 + (r + \frac{1}{2}\sigma^2)T$ and variance $\sigma^2 T$. So

$$P^S(S_T > K) = P\Big(Z > \frac{\ln(K/S_0) - (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\Big) = N(d_+).$$

And under $\tilde{P}$ it's $N(d_-)$, the same calculation with $r - \frac{1}{2}\sigma^2$. **d_+ and d_- differ only in the drift.**

PROBABILITY	SIMULATED	FORMULA	TERM IN C
$\tilde{P}(S_T > K)$	0.4230	$N(d_-) = 0.4231$	43.1133
$P^S(S_T > K)$	0.5019	$N(d_+) = 0.5024$	50.2413
$P(S_T > K)$, real world		0.5223	not used

Why $N(d_+)$ is bigger. In share units, paths on which the share ends high have more weight. Those are the paths on which the call finishes in the money. The gap here is 0.0793.

And delta. Since $\Delta = N(d_+)$, the hedge ratio is the probability of exercise **in share units**. That is the sense in which the old rule of thumb, "delta is the probability of finishing in the money", is right, once the unit is named.

A price with no rate in it

An **exchange option** pays $\max(S_T^1 - S_T^2, 0)$ at T: the right to swap one share of asset 2 for one of asset 1. Each asset has its own volatility, with correlation ρ between their Brownian motions. With cash as the numéraire, you'd face a two-dimensional integral. Count in shares of asset 2, and it collapses.

Step 1. Choose the numéraire. In units of asset 2, the payoff is $\max(S_T^1/S_T^2 - 1, 0)$. Let $Y = S^1/S^2$.

Step 2. Its motion. Y is a price counted in asset 2, so it's a martingale under P^{S^2}. Hence no drift, and just a volatility.

Step 3. Its volatility. Itô's lemma on the ratio gives

$$\frac{dY}{Y} = \sigma_1\, dW^1 - \sigma_2\, dW^2 + (\text{a } dt \text{ term that vanishes under } P^{S^2}).$$

The square of the random part, using $dW^1\, dW^2 = \rho\, dt$, gives

$$\sigma_x^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2.$$

Step 4. Read off Black-Scholes. $E^{S^2}[\max(Y_T - 1, 0)]$ is a Black-Scholes call on Y with strike 1, volatility σ_x and rate 0. Multiplying by S_0^2:

$$\boxed{V_0 = S_0^1 N(d_1) - S_0^2 N(d_2), \qquad d_{1,2} = \frac{\ln(S_0^1/S_0^2) \pm \frac{1}{2}\sigma_x^2 T}{\sigma_x\sqrt{T}}}$$

This is **Margrabe's formula**. **The interest rate r has gone.** Both assets grow at the same risk-neutral rate, and only the ratio matters.

Checked, with $S_0^1 = 100$, $S_0^2 = 95$, $\sigma_1 = 0.2$, $\sigma_2 = 0.3$, $\rho = 0.5$. Then $\sigma_x = 0.2646$, and the formula gives 12.9523.

Monte Carlo at three interest rates. On the same random numbers, the results agree to every digit: r cancels exactly.

RATE R, %	SIMULATED	STANDARD ERROR
0	12.9606	0.0110
3	12.9606	0.0110
10	12.9606	0.0110

What to remember

- A **numéraire** is a unit for counting wealth. Under its own probability, everything counted in that unit is a martingale.
- **Share measure:** weight $Z_T^S = e^{-rT} S_T / S_0$; drift rises by σ^2; volatility doesn't change.
- **Pricing in shares:** $V_0 = S_0 E^S[V_T / S_T]$.
- $C = S_0 P^S(S_T > K) - K e^{-rT} \tilde{P}(S_T > K)$. So $N(d_+)$ and $N(d_-)$ are the same event, in different units.
- $\Delta = N(d_+)$: delta is the probability of exercise in share units.
- **Margrabe:** count in asset 2, $\sigma_x^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, no r.

Common mistakes

- **Thinking the numéraire changes the price.** It changes how you compute it. The price is the same.
- **Mixing units.** Under P^S, divide by the share, don't discount by cash.
- **Using the $\tilde{P}$ drift under P^S.** It's $r + \sigma^2$ there.
- **Calling $N(d_+)$ "the probability of exercise" with no unit.** The real-world, cash and share probabilities are all different: here 0.522, 0.423 and 0.502.
- **Adding the two volatilities in Margrabe.** With $\rho = 1$ and equal volatilities, the ratio doesn't move at all. It's a difference, not a sum.

Exercises

Solutions are in *Part 4: Solutions*.

27.1 Show that $Z_T^S = e^{-rT} S_T / S_0$ has $\tilde{E}[Z_T^S] = 1$, in two ways: by the martingale property of the discounted share, and by writing it as an exponential of a normal.

27.2 Find the law of S_T under P^S. Using the MGF and the weight, show that $\ln S_T$ under P^S is normal with mean $\ln S_0 + (r + \frac{1}{2}\sigma^2)T$ and variance $\sigma^2 T$. What is the median of S_T under P^S with the course's numbers?

27.3 Use Itô's lemma on $Y_t = e^{rt}/S_t$ with $dS = (r + \sigma^2) S\, dt + \sigma S\, dW^S$. Show that the dt term vanishes, so cash counted in shares is a martingale under P^S.

27.4 Derive $P^S(S_T > K) = N(d_+)$ directly, by standardising $\ln S_T$ under P^S. Then write C as the sum of two terms, one priced with each unit, and explain in words why $N(d_+) > N(d_-)$.

27.5 Derive Margrabe's formula: count in units of asset 2, use the weight $Z = e^{-rT} S_T^2 / S_0^2$, find the ratio's volatility, and read the answer off the Black-Scholes formula with $r = 0$ and strike 1. Evaluate it for $S_0^1 = 100$, $S_0^2 = 95$, $\sigma_1 = 20\%$, $\sigma_2 = 30\%$, $\rho = 0.5$, $T = 1$.

27.6 Do the same split for the put, $P = K e^{-rT} N(-d_-) - S_0 N(-d_+)$. Read each $N(\cdot)$ as a probability in its own unit. Explain why the put's two probabilities are of finishing **below** the strike.

27.7 *(Code)* Estimate $N(d_+)$ two ways: by simulating under P^S directly, and by weighting $\tilde{P}$ paths with Z_T^S. Which has the smaller standard error? Why? Try $K = 105$ and $K = 160$.

27.8 *(Code)* Check Margrabe's formula by Monte Carlo for ρ of $-0.5, 0, 0.5, 0.9$, using correlated normals. How does the price depend on ρ, and why? Check too that changing r from 0 to 10% changes nothing.

Self-test

1. What is the drift of the share under P^S?
2. What does $N(d_+)$ measure, and what's its value with the course's numbers?
3. Why does Margrabe's formula contain no interest rate?

Companion notebook: `s5e27_numeraire.ipynb`. Every number in this chapter is computed by it.

LESSON 28 · VIDEO 28

The Feynman-Kac Theorem

An expectation as a function of the starting point; the theorem linking it to a PDE, proved in a few lines with the tower property and Itô's lemma; adding a discount; how Black-Scholes is a special case and what that says about μ; and bond prices under a moving interest rate.

Why this lesson

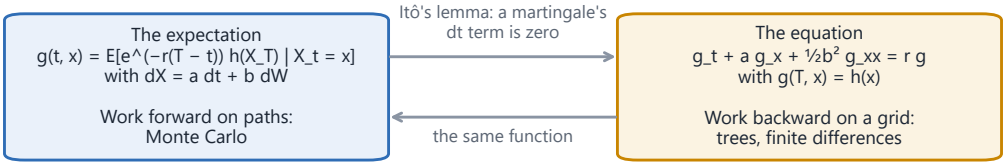
We've found the option's price in two ways that look unrelated: an **equation** (Lesson 19) and an **average** (Lesson 20). The two agreed, but you might fairly ask whether that's a coincidence.

The **Feynman-Kac theorem** says it isn't. Any expectation of this kind, as a function of the starting point, solves a PDE. And any PDE of this kind has a solution that's an expectation. This is one of the deepest links in applied mathematics. It's also deeply practical: it means every pricing problem can be attacked from either side, with paths or with grids. Lessons 29 and 30 use both.

What you need from earlier lessons

- **The tower property** (Lesson 8). $E[E[X \mid \mathcal{F}_t] \mid \mathcal{F}_s] = E[X \mid \mathcal{F}_s]$ for $s < t$.
- **The martingale test** (Lesson 18). A process $dY = a\,dt + b\,dW$ that's a martingale has $a = 0$.
- **Itô's lemma** (Lesson 16). $dg(t, X) = (g_t + ag_x + \frac{1}{2}b^2 g_{xx})\,dt + bg_x\,dW$.
- **The Black-Scholes equation** (Lesson 19), and **risk-neutral pricing** (Lesson 26).

The picture



For the Black-Scholes model, a = rS and b = σS: g is the option price, and the equation is Lesson 19's.

Figure 28.1. The two sides of Feynman-Kac. An expectation over future paths is the same function as the solution of a PDE. Each side gives its own numerical method.

An expectation that depends on where you start

The setup. Let X follow a **stochastic differential equation**:

$$dX_t = a(X_t)\,dt + b(X_t)\,dW_t.$$

The coefficients depend only on the current value. So where X goes next depends on where it is now, not on how it got there. This is the **Markov property**.

Fix a payoff h and a final time T. Define

$$g(t, x) = E\big[h(X_T) \mid X_t = x\big].$$

Start at x at time t, run to T, and average the payoff. Since X is Markov, this is an ordinary function of two numbers.

A toy example. Take $X = W$, $h(x) = x^2$. From x at t, $W_T = x + \sqrt{T-t}\,Z$, so $g(t, x) = x^2 + (T - t)$.

The expectation as a function of the starting point, with T = 1 and 1,000,000 paths.

T	X	SIMULATED	x² + (T − t)
0.0	0.0	1.0002	1.0000
0.0	1.0	1.9999	2.0000
0.5	1.0	1.4999	1.5000
0.5	-2.0	4.5006	4.5000
0.9	0.5	0.3500	0.3500

The theorem

> **Feynman-Kac.** Let $g(t,x) = E[h(X_T) \mid X_t = x]$ with $dX = a\,dt + b\,dW$. Then

$$g_t + a(x)\,g_x + \tfrac{1}{2}b(x)^2 g_{xx} = 0, \qquad g(T,x) = h(x).$$

Proof. It has only three steps, and each uses one earlier lesson.

Step 1. $g(t, X_t)$ is a martingale. By the Markov property, $g(t, X_t) = E[h(X_T) \mid \mathcal{F}_t]$. For $s < t$, by the tower property,

$$E[g(t,X_t) \mid \mathcal{F}_s] = E\big[E[h(X_T) \mid \mathcal{F}_t] \,\big|\, \mathcal{F}_s\big] = E[h(X_T) \mid \mathcal{F}_s] = g(s, X_s).$$

Intuitively: your best forecast today of tomorrow's forecast is today's forecast.

Step 2. Itô's lemma.

$$dg(t,X_t) = \big(g_t + a g_x + \tfrac{1}{2}b^2 g_{xx}\big)\,dt + b g_x\,dW.$$

Step 3. A martingale has no dt term (Lesson 18). So $g_t + a g_x + \frac{1}{2}b^2 g_{xx} = 0$ at every point the process can reach. ■

(We assumed that g is smooth enough, and that integrals are well-behaved. For smooth coefficients and sensible payoffs, these hold.)

The final condition. At $t = T$ there's nothing left to average: $g(T,x) = h(x)$.

Checked. For three cases, the left side of the equation evaluated by small bumps on a grid of (t,x):

Each known expectation satisfies its equation, up to the rounding of the bumps.

PROCESS AND PAYOFF	LARGEST RESIDUAL
W, payoff x^2	5.8e-10
W, payoff e^x	2.8e-07
GBM at 8%, payoff S	1.3e-11

The martingale, simulated. For $X = W$, $h = x^2$: $g(t, W_t) = W_t^2 + (T - t)$ should have a constant average, while its spread grows as time passes.

A martingale: the same average at every time, with more and more spread.

T	Average of g(t, W_t)	SD
0.25	0.9991	0.352
0.50	0.9986	0.704
0.75	0.9994	1.059
1.00	0.9967	1.406

The reverse direction. It works both ways. Suppose g solves the PDE with the final condition. Then Itô's lemma shows that $g(t, X_t)$ has no dt term, and so (given integrability) it's a martingale. Therefore $g(t,x) = E[g(T, X_T) \mid X_t = x] = E[h(X_T) \mid X_t = x]$. **A solution of the PDE is an expectation.**

Adding a discount

For pricing, we need a discounted payoff. Let r be constant, and

$$g(t,x) = E\big[e^{-r(T-t)} h(X_T) \mid X_t = x\big].$$

Now $g(t, X_t)$ itself isn't a martingale. But $e^{-rt} g(t, X_t) = e^{-rT} E[h(X_T) \mid \mathcal{F}_t]$ is, by the same tower argument. Apply the product rule and Itô:

$$d\big(e^{-rt} g\big) = e^{-rt}\big(g_t + a g_x + \tfrac{1}{2} b^2 g_{xx} - r g\big)\, dt + e^{-rt} b g_x \, dW.$$

Set the dt term to zero:

$$\boxed{g_t + a\, g_x + \tfrac{1}{2} b^2 g_{xx} = r g, \qquad g(T,x) = h(x)}$$

The discount adds the single term rg. **Checked** for a share at drift r with payoff S^2, where Lesson 19 gave the answer $S^2 e^{(r+\sigma^2)(T-t)}$:

A discounted expectation, on 1,000,000 paths.

T	S	SIMULATED	FORMULA
0.0	100	10725.2	10725.1
0.5	100	10356.0	10356.2
0.5	120	14912.7	14912.9

Back to Black-Scholes

Take $a(S) = rS$, $b(S) = \sigma S$, and the discounted version. Then the equation is

$$g_t + rS g_S + \tfrac{1}{2}\sigma^2 S^2 g_{SS} = r g,$$

which is the Black-Scholes equation, with h the option's payoff. And the expectation is the risk-neutral price of Lesson 26. Two different roads reach the same function, and the theorem is the bridge.

The puzzle of μ. In Lesson 19, μ cancelled. Feynman-Kac shows precisely why it's the $\tilde{P}$ expectation that appears. Compare two functions:

- $g_r(t, S) = \tilde{E}[e^{-r(T-t)} \max(S_T - K, 0) \mid S_t = S]$, with the share drifting at r. Value at the start: 7.1281.
- $g_\mu(t, S) = E[e^{-r(T-t)} \max(S_T - K, 0) \mid S_t = S]$, with the share drifting at μ. Value at the start: 9.9603.

Each is a perfectly good expectation, and each solves *its own* PDE. By Feynman-Kac, g_μ solves the equation with μ in the S-term.

	EQUATION WITH r IN THE S-TERM	EQUATION WITH μ IN THE S-TERM
Residual of g_r at $S = 100$, $t = 0.5$	2.2e-07	
Residual of g_μ at $S = 100$, $t = 0.5$	2.5808	2.7e-07

The resolution. The hedging argument of Lesson 19 selects the equation with r. Plugging g_μ into that equation leaves a remainder of 2.5808, which is exactly $(\mu - r)Sg_S = 2.5808$. **Feynman-Kac alone can't tell you which expectation is the price. Arbitrage does.** The real-world expectation is a true statement about the future, but it isn't what a hedged option costs.

A rate that moves: bonds under Vasicek

So far the discount has been constant. What if the short rate is random? Here is a preview of interest-rate models. Let

$$dr_t = \kappa(\bar{r} - r_t)\, dt + s\, dW_t,$$

the **Vasicek model**: the rate is pulled towards a long-run level $\bar{r}$ at speed κ, with some noise. A zero-coupon bond pays 1 at T:

$$P(t, r) = \tilde{E}\left[e^{-\int_t^T r_u\, du} \,\middle|\, r_t = r \right].$$

Feynman-Kac with a moving discount. The same proof goes through, with $e^{-\int_0^t r_u\,du}P(t,r_t)$ as the martingale. The only change is that the constant r in rg becomes the variable r:

$$P_t + \kappa(\bar{r} - r)P_r + \tfrac{1}{2}s^2 P_{rr} = rP, \qquad P(T,r) = 1.$$

Solving it. Try the form $P = e^{A(\tau) - B(\tau)r}$, with $\tau = T - t$. Substituting and collecting the terms in r gives ordinary differential equations, solved in Exercise 28.6:

$$B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}, \qquad A(\tau) = \Big(\bar{r} - \frac{s^2}{2\kappa^2}\Big)(B - \tau) - \frac{s^2 B^2}{4\kappa}.$$

Checked with $\kappa = 0.5$, $\bar{r} = 0.04$, $s = 0.01$, $r_0 = 0.03$:

Vasicek bond prices, by Feynman-Kac's formula and by simulating the rate. The yield rises from 3% towards the long-run level, as it should when the rate starts below it.

MATURITY	A	B	FORMULA	SIMULATED	STANDARD ERROR	YIELD, %
1	-0.00851	0.7869	0.96839	0.96838	0.00001	3.212
2	-0.02936	1.2642	0.93492	0.93493	0.00003	3.365
5	-0.12610	1.8358	0.83429	0.83425	0.00008	3.624
10	-0.31913	1.9865	0.68473	0.68472	0.00012	3.787

The PDE's residual at r_0 at maturity 5 is 3.7e-09. Interest-rate models are built on exactly this calculation.

What to remember

- **Markov:** $dX = a(X)\,dt + b(X)\,dW$; the future depends only on the present.
- **Feynman-Kac:** $g = E[h(X_T) \mid X_t = x]$ solves $g_t + ag_x + \tfrac{1}{2}b^2 g_{xx} = 0$, $g(T,x) = h(x)$.
- **The proof:** tower property $\Rightarrow$ martingale; Itô $\Rightarrow$ dt term; martingale $\Rightarrow$ dt term is zero.
- **With a discount:** the right side becomes rg. If r is a random state variable, it's the variable r times g.
- **Black-Scholes** is the case $a = rS$, $b = \sigma S$. The $\tilde{P}$ expectation is what arbitrage selects.
- It works both ways, so each pricing problem can be solved with paths or with a grid.

Common mistakes

- **Using Feynman-Kac to justify real-world pricing.** The theorem is about mathematics; arbitrage chooses the drift.
- **Forgetting the terminal condition.** The PDE alone has infinitely many solutions. The payoff picks one.
- **Applying it to non-Markov processes.** If the coefficients depend on the whole past, there's no function of (t, x) alone.
- **Sign of the discount term.** It's $= rg$, or equivalently $g_t + \cdots - rg = 0$.
- **Treating τ and t alike.** With $\tau = T - t$, $g_t = -g_\tau$.

Exercises

Solutions are in *Part 4: Solutions*.

28.1 Show that $g(t, X_t) = E[h(X_T) \mid \mathcal{F}_t]$ is a martingale, using the tower property. Where does the Markov property come in, in saying that this conditional expectation is a function of X_t alone?

28.2 Using Feynman-Kac with $X = W$, write down the equation and terminal condition for $h(x) = x^2$ and for $h(x) = e^x$. Solve each by guessing a form, $x^2 + c(t)$ and $e^{x + c(t)}$, and check the answers against direct expectations.

28.3 Prove the discounted version: show that $e^{-rt} g(t, X_t)$ is a martingale, apply Itô's lemma, and set the dt term to zero.

28.4 Let S be a GBM with drift μ and $h(S) = S$. Write the equation for $g = E[S_T \mid S_t = S]$, guess $g = Sc(t)$, and solve. Then do the same with the discount, and show the discounted price is S when $\mu = r$.

28.5 A friend says: "Feynman-Kac shows the option's value is an expectation under the real probability, with μ." Explain precisely what is wrong, by writing out which expectation solves which equation. What is the extra term that a real-world expectation leaves in the Black-Scholes equation?

28.6 For Vasicek, substitute $P = e^{A(\tau) - B(\tau) r}$ into the bond equation, with $\tau = T - t$. Collect the terms with r and without. Solve for B, and write down the equation for A and its solution.

28.7 *(Code)* Use Feynman-Kac with $X = W$ and $h(x) = x^4$: guess the polynomial $g = x^4 + c_1(t) x^2 + c_2(t)$, solve for c_1 and c_2 by hand, and confirm by simulation at several (t, x) and by computing the PDE residual with bumps.

28.8 *(Code)* Price Vasicek bonds by Monte Carlo at maturities 1, 2, 5, 10 with the lesson's parameters. Compare the formula, and find how the error depends on the number of time steps per year (10, 50, 250). Use both a left-endpoint sum and a trapezium sum for $\int r\,dt$.

Self-test

1. For $X = W$, $h(x) = x^2$, $T = 1$: what is $g(0.5, 1)$?
2. Which two facts force the dt term of $g(t, X_t)$ to vanish?
3. Which term does a constant discount rate add to the equation?

Companion notebook: `s5e28_feynman_kac.ipynb`. Every number in this chapter is computed by it.

LESSON 29 · VIDEO 29

Finite Differences

Replacing derivatives with differences, and the size of the error each makes; the Black-Scholes equation in log price; the explicit scheme, which is a trinomial tree on a fixed grid; the stability limit and what breaking it looks like; and the implicit and Crank-Nicolson schemes, which remove the limit.

Why this lesson

Feynman-Kac (Lesson 28) gives two ways to attack a pricing problem. Monte Carlo works with the expectation. **Finite differences** work with the PDE: lay a grid over prices and times, replace every derivative with a difference of neighbouring values, and march from the payoff back to today.

It's the workhorse for low-dimensional problems. It gives the whole price curve at once, handles early exercise in one line, and yields Greeks for free. It also has a famous trap: choose the steps badly, and your solution blows up to 10^{30}. This lesson derives the scheme, measures its accuracy, shows exactly why it blows up and at what point, and then removes the trap.

What you need from earlier lessons

- **Taylor's theorem** (Lesson 1). $f(x+h) = f(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + \frac{1}{6}h^3 f'''(x) + \dots$
- **The Black-Scholes equation** (Lesson 19). $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$.
- **The trinomial tree** (Lesson 23). Three branches, moment matching, the strike on a node.
- **The log price** (Lesson 17). $\nu = r - \frac{1}{2}\sigma^2$.

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$, so $\nu = 0.01$. Call 7.1281.

The picture

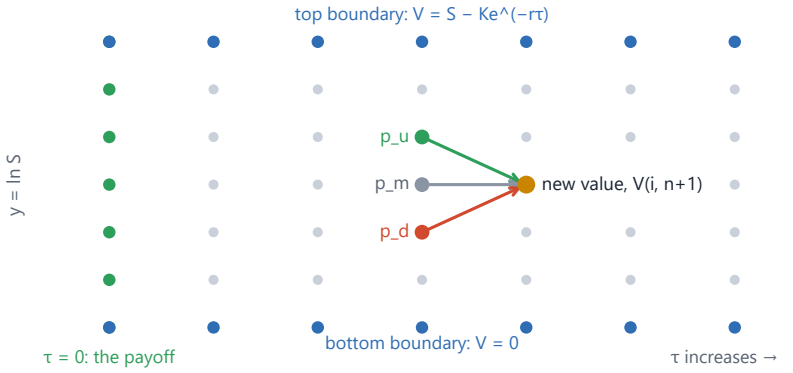


Figure 29.1. The grid. Values at τ = 0 are the payoff (green). Top and bottom edges are boundary conditions (blue). Each new value comes from three neighbours at the previous time, weighted by p_u, p_m, p_d.

The stencil looks exactly like a trinomial tree's three branches. That's no coincidence.

Replacing derivatives with differences

Take a smooth function and a small spacing h. By Taylor's theorem:

$$f(x+h) = f(x) + hf'(x) + \tfrac{1}{2}h^2 f''(x) + \tfrac{1}{6}h^3 f'''(x) + \dots$$

$$f(x-h) = f(x) - hf'(x) + \tfrac{1}{2}h^2 f''(x) - \tfrac{1}{6}h^3 f'''(x) + \dots$$

Forward difference. Rearranging the first line:

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \tfrac{1}{2}hf''(x) + \dots$$

The error is proportional to h: **first order**.

Central difference. Subtract the two lines: the even terms cancel.

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \tfrac{1}{6}h^2 f'''(x) + \dots$$

The error is proportional to h^2: **second order**.

Second difference. Add the two lines: the odd terms cancel.

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \tfrac{1}{12}h^2 f''''(x) + \dots$$

Again second order.

Checked, on $f = e^x$ at 0, where every derivative is 1:

Halving h halves the forward error (ratio 2.02) and quarters the other two (ratio 4.00).

H	FORWARD ERROR	CENTRAL ERROR	SECOND-DIFFERENCE ERROR
0.200	0.107014	0.006680	0.003338
0.100	0.051709	0.001668	0.000834
0.050	0.025422	0.000417	0.000208
0.025	0.012605	0.000104	0.000052

The equation in log price

As in Lesson 23, we'd like constant coefficients, so that the same weights work at every grid point. Change variables to $y = \ln S$ and $\tau = T - t$, so $V(t, S) = U(\tau, y)$.

Step 1. The derivatives, by the chain rule. $V_t = -U_\tau$. $V_S = U_y / S$. And $V_{SS} = (U_{yy} - U_y)/S^2$. (For the last one: $\partial_S (U_y/S) = U_{yy}/S^2 - U_y/S^2$.)

Step 2. Substitute. The factors of S cancel:

$$-U_\tau + rU_y + \tfrac{1}{2}\sigma^2 (U_{yy} - U_y) = rU.$$

Step 3. Rearrange, with $\nu = r - \tfrac{1}{2}\sigma^2$:

$$\boxed{U_\tau = \nu U_y + \tfrac{1}{2}\sigma^2 U_{yy} - rU, \qquad U(0, y) = \max(e^y - K, 0)}$$

A constant-coefficient equation, starting from the payoff at $\tau = 0$, moving forward in τ. **Checked** on the formula at $S = 100$, $\tau = 0.5$: the left side is 6.7369, and the right side is 6.7369.

The explicit scheme

The grid. Let $y_i = \ln S_0 + i\Delta y$ for i from $-M/2$ to $M/2$, and $\tau_n = n\Delta\tau$, with $N\Delta\tau = T$. Write U_i^n for the value at grid point (i, n).

Discretise. Use a forward difference in τ, and central differences in y, all at the **old** time n. For the discount term, we use the new time, which makes the formula tidier:

$$\frac{U_i^{n+1} - U_i^n}{\Delta\tau} = \nu \frac{U_{i+1}^n - U_{i-1}^n}{2\Delta y} + \frac{1}{2}\sigma^2 \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta y^2} - rU_i^{n+1}.$$

Solve for the new value. Multiply by $\Delta\tau$, move terms and collect by neighbour:

$$\boxed{U_i^{n+1} = \frac{p_u U_{i+1}^n + p_m U_i^n + p_d U_{i-1}^n}{1 + r\Delta\tau}}$$

$$p_u = \Delta\tau\Big(\frac{\sigma^2}{2\Delta y^2} + \frac{\nu}{2\Delta y}\Big), \quad p_m = 1 - \frac{\sigma^2\Delta\tau}{\Delta y^2}, \quad p_d = \Delta\tau\Big(\frac{\sigma^2}{2\Delta y^2} - \frac{\nu}{2\Delta y}\Big).$$

Each new value is computed **explicitly** from known old ones.

It's a trinomial tree. The three weights sum to 1. So each new value is a discounted average of three neighbours, just like a tree. Set $\Delta y = \sigma\sqrt{3\Delta\tau}$ and $\Delta\tau = 0.01$, and the weights are 0.16811, 0.66667 and 0.16522: Lesson 23's probabilities, exactly. Feynman-Kac's two sides meet here: the probabilities of the paths are the coefficients of the PDE.

The boundaries. The stencil needs neighbours that don't exist at the grid's edges. Take the grid far enough out ($\pm 5\sigma\sqrt{T}$ in y, which is share prices from about 37 to about 272), and use what's known there: at the bottom a call is worth 0, and at the top it's worth $S - Ke^{-r\tau}$.

Accuracy.

Refining the grid, with Δτ falling as Δy². As with the trees, the raw error is untidy. With the strike on a grid point, it settles down to quartering each time.

M	N	σ²Δτ/Δy²	CALL	ERROR	ERROR, STRIKE ON A GRID POINT
100	1,000	0.10	7.1294	0.00134	-0.01303
200	4,000	0.10	7.1270	-0.00106	-0.00201
400	16,000	0.10	7.1280	-0.00007	-0.00050
800	64,000	0.10	7.1281	0.00002	-0.00013

Reading the table. The second-to-last column is untidy, for the same reason as Lesson 22: the strike sits at a different place between grid points. Nudge the spacing so the strike is on a grid point, and from $M = 200$ on the last column falls by 4.00 each time M doubles and N quadruples. That's second-order accuracy in Δy.

When it blows up

Now the trap. In Lesson 23, p_m was a probability. Here, $p_m = 1 - \sigma^2\Delta\tau/\Delta y^2$ can go negative if the time step is too big.

The ratio that matters is $c = \sigma^2\Delta\tau/\Delta y^2$. The scheme is an average, and so can't amplify anything, only while $c \leq 1$.

How the error grows. Take $\nu = r = 0$, and suppose the error at step n is a zigzag: $+\varepsilon, -\varepsilon, +\varepsilon, \ldots$, the fastest wiggle the grid can show. Then with $p_u = p_d = c/2$ and $p_m = 1 - c$, each point's new error is

$$(1 - c)(\pm\varepsilon) + c(\mp\varepsilon) = \pm(1 - 2c)\varepsilon.$$

So each step multiplies the zigzag by $1 - 2c$. If $c \leq 1$, its size is at most 1, and the error dies away. If $c > 1$, it exceeds 1 in size, and the zigzag **grows exponentially, step after step**. Round-off, always present, is enough to start it. Exercise 29.4 shows this is the worst case among all wiggles.

In numbers. With $M = 200$ and $\Delta y = 0.01$, the limit is $N = \sigma^2 T/\Delta y^2 = 400$.

M = 200. The scheme works all the way to c = 1. Just beyond that, it fails completely.

TIME STEPS N	C	P_M	WORST GROWTH PER STEP	CALL PRICE
2,000	0.200	0.800	0.600	7.1272
800	0.500	0.500	0.000	7.1278
450	0.889	0.111	0.778	7.1285
400	1.000	0.000	1.000	7.1325
395	1.013	-0.013	1.025	blown up
380	1.053	-0.053	1.105	blown up
300	1.333	-0.333	1.667	blown up

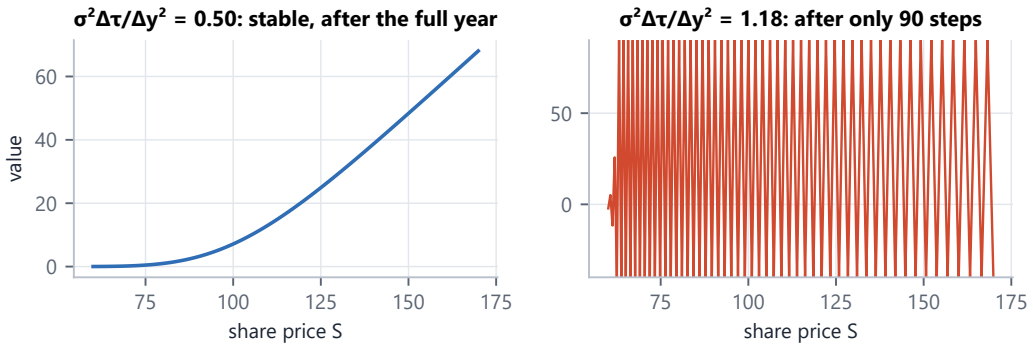


Figure 29.2. Left: a stable run. Right: the same grid at c = 1.18, after only 90 of its 340 steps. The zigzag is already huge, and the values were clipped for plotting; the largest is about 6×10^9.

The cost of stability. Halving Δy forces $\Delta\tau$ down by 4. So the total work grows as $M \times N \propto M^3$. And what does that buy? The error falls as M^{-2}. So work grows like $\varepsilon^{-3/2}$. Better than Monte Carlo's ε^{-2}, but the time-step limit is annoying.

The implicit scheme

The idea. Use the differences at the **new** time:

$$\frac{U_i^{n+1} - U_i^n}{\Delta\tau} = \nu\frac{U_{i+1}^{n+1} - U_{i-1}^{n+1}}{2\Delta y} + \frac{1}{2}\sigma^2\frac{U_{i+1}^{n+1} - 2U_i^{n+1} + U_{i-1}^{n+1}}{\Delta y^2} - rU_i^{n+1}.$$

Now the new values appear on both sides. Each row links three unknowns, so each time step means solving a **tridiagonal** system of equations:

$$-\Delta\tau\, a_d\, U_{i-1}^{n+1} + (1 - \Delta\tau\, a_m)U_i^{n+1} - \Delta\tau\, a_u\, U_{i+1}^{n+1} = U_i^n,$$

with $a_u = \frac{\sigma^2}{2\Delta y^2} + \frac{\nu}{2\Delta y}$, $a_m = -\frac{\sigma^2}{\Delta y^2} - r$, and $a_d = \frac{\sigma^2}{2\Delta y^2} - \frac{\nu}{2\Delta y}$.

It's cheap. A tridiagonal system is solved by one forward sweep and one backward sweep (the **Thomas algorithm**), about $3M$ operations, hardly more than an explicit step.

It doesn't blow up, whatever the time step. For the zigzag, the growth factor is $1/(1 + 2c)$, always less than 1.

Crank-Nicolson averages the two: half old, half new. It's still a tridiagonal solve, its zigzag factor is $(1 - c)/(1 + c)$, and its time error is second-order rather than first.

The comparison. $M = 200$, errors against the formula:

The errors of the three schemes. The implicit scheme's falls by about 2 when N doubles. Crank-Nicolson has already reached the grid's spacing error, about −0.00125, even at N = 50.

N	C	EXPLICIT	IMPLICIT	CRANK-NICOLSON
50	8.00	blown up	-0.01985	-0.00117
100	4.00	blown up	-0.01056	-0.00123
200	2.00	blown up	-0.00591	-0.00125
400	1.00	0.00446	-0.00358	-0.00125
1,000	0.40	-0.00048	-0.00219	-0.00125

Reading the table. The explicit scheme is useless below $N = 400$. The implicit scheme gives sensible answers anywhere, with first-order time error. Crank-Nicolson is the most accurate of the three at low N. (It has its own weakness: at very large c, its factor $(1 - c)/(1 + c)$ approaches -1, so a zigzag from the payoff's kink dies out only slowly. Banks often start with a few implicit steps.)

Early exercise. On any of these grids, add the check of Lesson 22 after each step: replace U_i with the larger of U_i and the exercise value. (For implicit schemes it's slightly more subtle, as the check should be part of the solve.) On the explicit grid with $(M, N) = (400, 16000)$, the American put comes to 9.477, in agreement with Lesson 22's tree, and the European put to 9.0248 (formula 9.0248).

What to remember

- **Differences:** forward has error $O(h)$; central and second have $O(h^2)$.
- **Log variables:** $U_\tau = \nu U_y + \frac{1}{2}\sigma^2 U_{yy} - rU$. Constant coefficients.
- **Explicit:** $U_i^{n+1} = (p_u U_{i+1}^n + p_m U_i^n + p_d U_{i-1}^n)/(1 + r\Delta\tau)$: a trinomial tree on a fixed grid.
- **Stability:** $c = \sigma^2 \Delta\tau / \Delta y^2 \le 1$. Beyond that, a zigzag grows by $|1 - 2c|$ per step.
- **Implicit:** a tridiagonal solve per step; no stability limit; first-order in time.
- **Crank-Nicolson:** the average; second-order in time; slow to damp a kink at very large c.
- Accuracy tidies up when the strike is on a grid point.

Common mistakes

- **Choosing Δy and $\Delta\tau$ independently in an explicit scheme.** Check $c \le 1$.
- **Refining only in space.** Halving Δy with $\Delta\tau$ fixed makes things worse: c goes up by 4.
- **Grid too narrow.** With boundaries only $\pm 1\sigma\sqrt{T}$ out, wrong boundary values leak into the answer.
- **Forgetting the τ in the top boundary.** It's $S - Ke^{-r\tau}$, not $S - K$.
- **Reading a price off a grid without interpolating.** If S_0 isn't on a grid point, interpolate (or centre the grid on S_0, as here).

Exercises

Solutions are in *Part 4: Solutions*.

29.1 Use Taylor's theorem to show that the central difference $\frac{f(x+h) - f(x-h)}{2h}$ has error $\frac{1}{6}f'''(x)h^2 + \ldots$. Compare with the forward difference's error. Check the ratio with $f = e^x$, $h = 0.1$.

29.2 Do the same for the second difference: show that $\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f'' + \frac{1}{12}f''''h^2 + \ldots$.

29.3 Starting from $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$, change variables to $y = \ln S$ and $\tau = T - t$. Derive the log equation. Then discretise it to get the explicit scheme, including the factor $1/(1 + r\Delta\tau)$. Show that the three weights add to 1.

29.4 Stability. Try a wave $V_i^n = A^n e^{\mathrm{i}i\xi\Delta y}$ in the scheme with $\nu = r = 0$, where $\mathrm{i}^2 = -1$. Show $A = 1 - 2c\sin^2(\xi\Delta y/2)$, with $c = \sigma^2\Delta\tau/\Delta y^2$. For which c is $|A| \le 1$ at every wave? Explain the result by how an average of neighbours behaves.

29.5 Halving Δy forces $\Delta\tau$ down by 4 to stay stable. If the space error is $O(\Delta y^2)$, how does the total work grow as the error falls? Compare with a Monte Carlo error of $O(\sqrt{1/\text{work}})$: which one wins, and at what error, for one share?

29.6 Why do we need boundary values at the grid's two ends? Justify $V = 0$ at the bottom and $V = S - Ke^{-r\tau}$ at the top for a call. Write the corresponding values for a put. How far out must the grid go for the choice not to matter? (Compare with the standard deviation of $\ln S_T$.)

29.7 *(Code)* Use the explicit scheme to price the European put, then add an early-exercise check for the American put, on grids (M, N) of $(100, 1000)$, $(200, 4000)$, $(400, 16000)$. Compare with the formula and with Lesson 22's tree value 9.478. What is the error's order?

29.8 *(Code)* With $M = 200$, compare the explicit, implicit and Crank-Nicolson schemes at $N = 50, 100, 200, 400, 1000$. Which give sensible answers when the explicit limit is broken? Estimate the time-order of the implicit and Crank-Nicolson errors, keeping $M = 800$ fixed.

Self-test

1. With $M = 200$ on our grid ($\Delta y = 0.01$), what's the smallest N that keeps the explicit scheme stable?
2. Which difference formula has error proportional to h, and which to h^2?
3. What do you have to do at each time step of an implicit scheme, and why is it cheap?

Companion notebook: `s5e29_finite_differences.ipynb` . Every number in this chapter is computed by it.

LESSON 30 · VIDEO 30

Choosing a Method

Feynman-Kac's two sides as a map of numerical methods; one call priced every way; early exercise and Longstaff-Schwartz; why a grid can't handle five shares; path-dependent payoffs; Greeks by bumping, pathwise and likelihood ratio; and a decision chart that sums up the course.

Why this lesson

You now own a full toolbox. The formula, binomial and trinomial trees, explicit and implicit grids, and Monte Carlo all give the same price where they overlap. Which should you use, and when?

The short answer comes from Feynman-Kac (Lesson 28), which divides all methods in two. **The PDE side** works backwards on a grid: trees and finite differences. **The expectation side** works forward on paths: Monte Carlo. Each has a natural strength, and each stumbles where the other walks. This final lesson of Part 4 compares them on the problems that tell them apart: one option priced every way, then early exercise, many assets, path dependence and Greeks.

What you need from earlier lessons

- **Monte Carlo and its error bar** (Lesson 22), **trees** (Lessons 22 and 23), **grids** (Lesson 29).
- **Feynman-Kac** (Lesson 28), and **the likelihood weights** of Lesson 25.
- **The Greeks** (Lesson 21).
- **Correlated normals** (Lesson 14).

Numbers. $S_0 = 100$, $K = 105$, $r = 3\%$, $\sigma = 20\%$, $T = 1$. The call is 7.1281.

The picture

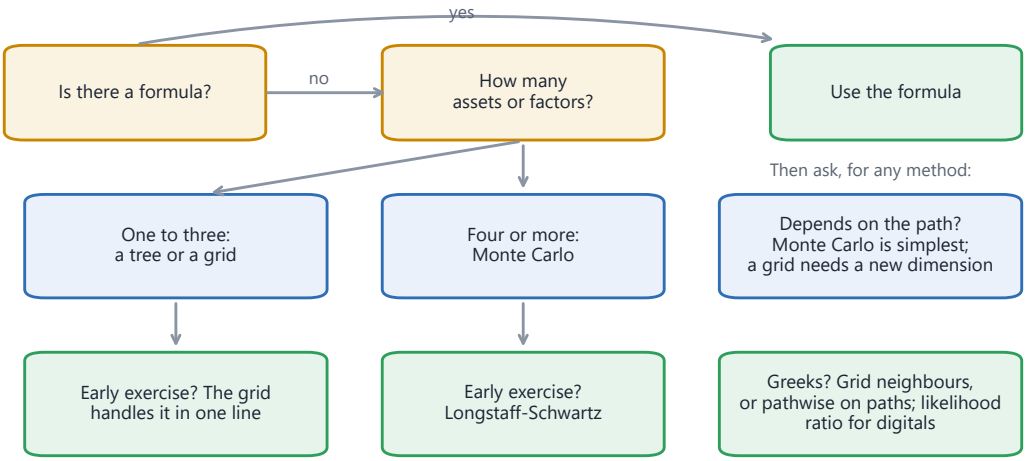


Figure 30.1. A way to choose. A formula first, if there is one. Then count the assets or factors: up to about three, a grid. More, Monte Carlo. Early exercise, path dependence and Greeks then refine the choice.

No chart replaces judgment, but this one is right most of the time. The rest of the lesson justifies each box.

One option, every method

Same call, all methods. Work is counted in node updates or paths, not in seconds, so the comparison holds on any machine.

One call, every method. The formula uses one evaluation. Monte Carlo's standard error here is about 0.013, so its small error in this run is luck.

METHOD	SIDE	PRICE	ERROR	WORK
Formula	both	7.1281	0.00000	1
Binomial tree, 500 steps	PDE	7.1301	0.00207	125,250
Trinomial, strike on a node, 450 steps	PDE	7.1237	-0.00441	203,401
Explicit differences, 200 x 800	PDE	7.1278	-0.00029	160,000
Monte Carlo, 1,000,000 paths	paths	7.1282	0.00015	1,000,000

Three things stand out.

- **With a formula, use it.** It's exact and instant, and it's your test case for everything else.

- **On one share, a grid beats paths.** Between 100,000 and 200,000 updates gets a tree or grid to within about 0.004, or much closer. A million Monte Carlo paths has a standard error of 0.013.
- **Paths give an honest error bar on their own.** The tree's and the grid's errors are known above only because there's a formula to compare with.

Early exercise

An American put must decide at each moment whether exercising beats continuing.

On a tree or grid the continuation value is right there at each node, so it's one line: $\max(\text{continue}, \text{exercise})$. With a large tree, the value is 9.4776, a premium of 0.453 over the European put's 9.0248.

On paths it's hard. Standing at a node on one path, you can't see the average future without looking at other paths. **Longstaff-Schwartz** (2001) gets an estimate by regression:

Step 1. Simulate all the paths forward.

Step 2. Work backwards from expiry, keeping each path's cash flow, discounted to the current date.

Step 3. At each date, take the paths in the money. Regress their discounted future cash flows on the current share price, using a polynomial. The fitted curve estimates the continuation value.

Step 4. On each such path, if exercise beats the fitted value, exercise: replace the path's cash flow with the exercise value.

Step 5. Average the discounted cash flows at the start.

Longstaff-Schwartz with a cubic. The tree allows exercise at any time.

PATHS	EXERCISE DATES	PRICE	STANDARD ERROR	GAP FROM THE TREE
20,000	12	9.472	0.065	-0.005
200,000	12	9.465	0.021	-0.013
200,000	50	9.455	0.020	-0.023
200,000	252	9.444	0.020	-0.034

Reading the table. Every estimate is within about two standard errors of the tree, and they tend to fall below it. There are two reasons. Fewer exercise dates are worth less. And an estimated rule can't beat the best one: following any actual rule gives a low esti-

mate (Exercises 30.6 and 30.7). For one share, it's a poor choice next to a tree. Its value is for many shares, where no tree is possible.

Many assets

A grid needs a point for each combination of share prices. With 100 points per share:

The curse of dimensionality. Grid size is exponential in the number of assets; Monte Carlo's work is linear.

ASSETS	GRID POINTS	MEMORY, BYTES	Monte Carlo draws, 10^6 paths
1	1.0e+02	8.0e+02	1,000,000
2	1.0e+04	8.0e+04	2,000,000
3	1.0e+06	8.0e+06	3,000,000
4	1.0e+08	8.0e+08	4,000,000
5	1.0e+10	8.0e+10	5,000,000
10	1.0e+20	8.0e+20	10,000,000

At 5 assets, the grid needs 80 gigabytes for one time step. At 10, more memory than exists. This is the **curse of dimensionality**. And Monte Carlo's error doesn't care: it's $s/\sqrt{M}$ at any dimension.

A five-share basket call, paying $\max(\text{average of the five} - K, 0)$, each with $\sigma = 20\%$ and pairwise correlation 0.5, on 1,000,000 paths: 5.339, with a standard error of 0.009.

Diversification lowers the basket's volatility, to about 0.155. A Black-Scholes call with that volatility (matching the basket's first two moments) gives 5.334, a useful check. A single share's call is 7.128.

Payoffs that depend on the path

An **Asian** call pays on the average price over the option's life, rather than the final price. On paths, it's one line of code: average along each path. On a grid, it's hard, since the running average is a second state variable, so you need a whole extra dimension.

200,000 paths. The more often the price is sampled, the lower the average's volatility, and so the option's value.

AVERAGING DATES	ASIAN CALL	STANDARD ERROR
12	3.474	0.015
52	3.231	0.014
252	3.165	0.014

The Asian call is worth 0.44 of the European one, because an average moves far less than a final price. **Paths handle path dependence naturally.** Barriers, lookbacks and autocallables are the same story, though a one-share barrier is also fine on a grid.

Greeks from each method

On a grid or tree, the Greeks are nearly free. Delta and gamma come from neighbouring nodes, and theta from neighbouring times. With a 1000-step tree and a bump of 1: delta 0.5011, against the formula's 0.5024.

On paths, there are three common ways, and it's worth seeing all of them. Take the call's delta.

Bumping. Reprice at $S_0 \pm b$, and take the difference. The key is to use the **same random numbers** for both repricings. With different ones, you're subtracting two noisy numbers, each with an error much bigger than the difference.

Pathwise. Differentiate the payoff along each path, with the random number held fixed. Since S_T is proportional to S_0, $\partial S_T/\partial S_0 = S_T/S_0$, and so

$$\text{delta estimate} = e^{-rT}\mathbf{1}\{S_T > K\}\frac{S_T}{S_0}.$$

The average is $N(d_+)$ exactly (Exercise 30.4). It works because the payoff's kink is a corner, not a jump.

Likelihood ratio. Differentiate the *density* instead of the payoff, and keep the payoff as is (Exercise 30.5):

$$\text{delta estimate} = e^{-rT}\max(S_T - K, 0)\frac{Z}{S_0\sigma\sqrt{T}}.$$

The call's delta, on 1,000,000 paths. The formula gives 0.5024. Using different random numbers is 15 times noisier than using the same ones.

METHOD	DELTA	STANDARD ERROR
Bump, different random numbers	0.5023	0.0089
Bump, same random numbers	0.5023	0.0006
Pathwise	0.5024	0.0006
Likelihood ratio	0.5020	0.0013

Why the likelihood ratio earns its place. For a **digital**, the payoff is a jump. The pathwise estimate is zero on every path, which is wrong. A bump has to catch the rare paths that land in the tiny window between the two strikes.

A cash digital's delta by bumping, on 1,000,000 paths. A big bump is biased; a small one is noisy.

BUMP	DELTA	STANDARD ERROR
5.00	0.01878	0.00004
1.00	0.01893	0.00009
0.10	0.01871	0.00030
0.01	0.02009	0.00099

The exact delta is 0.01900. The likelihood-ratio estimate is 0.01899, with a standard error of 0.00003: no bump to choose, no bias, and the smallest error of all.

What to remember

- **A formula first,** if there is one. It's also how you test everything else.
- **One to three assets:** a tree or grid. Fast, gives the Greeks for free, handles early exercise in one line.
- **Four or more:** Monte Carlo. Error $s/\sqrt{M}$ at any dimension. Honest error bar.
- **Early exercise on paths:** Longstaff-Schwartz; it gives a low estimate if the rule is valued on fresh paths.
- **Path dependence:** natural on paths; costs a dimension on a grid.
- **Greeks on paths:** bump with the same random numbers; pathwise for kinks; likelihood ratio for jumps.

Common mistakes

- **Reaching for Monte Carlo on a one-share European option.** It works, but a grid is thousands of times more efficient.
- **Building a four-dimensional grid.** Count the points first.
- **Bumping with fresh random numbers.** The noise swamps the signal.
- **Using a pathwise derivative on a digital.** It gives zero on every path.
- **Comparing methods by one lucky run.** Compare worst errors or standard errors.

Exercises

Solutions are in *Part 4: Solutions*.

30.1 A one-share tree has n steps with cost $\propto n^2$ and error $\propto 1/n$. A Monte Carlo price has M paths with cost $\propto M$ and error $\propto 1/\sqrt{M}$. How does the cost of each grow as the target error ε falls? Which wins at small ε, and what changes with more assets?

30.2 A grid has 100 points per asset. How many points, and how much memory at 8 bytes each, for $d = 1, 2, 3, 5, 10$? How many time steps would an explicit scheme need, given the stability limit? Roughly when is a grid impossible on a laptop?

30.3 For each of these, name a suitable method and say why: a European put; an American put on one share; a call on the average of 20 shares; a one-share barrier option that dies if the price touches 80; an American option on 5 shares; the delta of a digital on one share.

30.4 Derive the pathwise delta: differentiate $e^{-rT}\max(S_T - K, 0)$ along the path, with $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z}$ and Z fixed. Show that its expectation is $N(d_+)$. Why is it fine that the payoff has a kink?

30.5 Why does the pathwise method fail for a digital? Derive the likelihood-ratio formula: write the price as $\int f(s)\,p(s;S_0)\,ds$, differentiate the density, and show the weight is $Z/(S_0\sigma\sqrt{T})$.

30.6 In Longstaff-Schwartz, the exercise rule is fitted to the paths, and then the same paths value the option. Explain why, if a fresh set of paths is used for valuation, the answer must be a low estimate of the true American value. Why can reusing the same paths push it up?

30.7 *(Code)* Run Longstaff-Schwartz with polynomial degrees 1, 2, 3, 5 and exercise dates 12, 50. Use one set of paths to fit the rule and a separate set to value it. Compare with the tree. How much does each choice matter?

30.8 *(Code)* Compute the delta of the cash digital by bumping with same random numbers at bumps 5, 1, 0.1, 0.01, and by the likelihood-ratio formula, and compare with the exact $e^{-rT}\varphi(d_-)/(S_0\sigma\sqrt{T})$. Why does the bump's standard error rise as the bump shrinks? Which would you use?

Self-test

1. Why is a grid unusable for a call on ten shares?
2. Why must a bump use the same random numbers?
3. Which method would you use for the delta of a digital, on paths, and why not the pathwise one?

Companion notebook: `s5e30_choosing_a_method.ipynb`. Every number in this chapter is computed by it.

Formula sheet and glossary

Formula sheet

The equation (Lesson 19)

- Self-financing: $dX = \Delta\, dS + r(X - \Delta S)\, dt$.
- Hedge: $\Delta = V_S$.
- $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$. With a dividend yield q: $(r-q)SV_S$.

The formula (Lesson 20)

- $d_\pm = \dfrac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$; $d_+ = d_- + \sigma\sqrt{\tau}$.
- $C = SN(d_+) - Ke^{-r\tau}N(d_-)$; $P = Ke^{-r\tau}N(-d_-) - SN(-d_+)$.
- Cash digital: $e^{-r\tau}N(d_-)$. Asset digital: $SN(d_+)$.

The Greeks (Lesson 21)

- Identity: $S\varphi(d_+) = Ke^{-r\tau}\varphi(d_-)$.
- $\Delta = N(d_+)$; $\Gamma = \dfrac{\varphi(d_+)}{S\sigma\sqrt{\tau}}$; $\mathcal{V} = S\varphi(d_+)\sqrt{\tau} = \sigma\tau S^2\Gamma$.
- $\Theta = -\dfrac{S\varphi(d_+)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_-)$.
- Balance: $\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2\Gamma = rV$.

Numerical checks (Lesson 22)

- SE $= s/\sqrt{M}$; 95% interval $\bar{Y} \pm 1.96$ SE.
- Richardson: error c/n gives $V \approx 2V_{2n} - V_n$.
- Control variate: $\hat{V} = \bar{Y} - b(\bar{X} - E[X])$; variance falls by $1 - \rho^2$.

The trinomial tree (Lesson 23)

- $\Delta x = \lambda\sigma\sqrt{\Delta t}$; $p_m = 1 - 1/\lambda^2$; $p_{u,d} = \frac{1}{2}(1/\lambda^2 \pm \nu\Delta t/\Delta x)$; $\nu = r - \frac{1}{2}\sigma^2$.
- $\lambda = \sqrt{3}$ matches the fourth moment. $p_d \geq 0$ needs $\Delta t \leq \sigma^2/(\lambda^2\nu^2)$.

Hedging (Lesson 24)

- One step, seller: $\frac{1}{2}\Gamma S^2(\sigma^2\delta t - (\delta S/S)^2)$; break-even move $\sigma\sqrt{\delta t}$.

- Sd of final error $\propto 1/\sqrt{n}$; rule of thumb $\sqrt{\pi/4}\,\sigma\mathcal{V}/\sqrt{n}$.
- Wrong volatility, mean present value: $\frac{1}{2}(\sigma^2 - \sigma^2_{\text{true}}) \int_0^T e^{-rt} E[S^2\Gamma]\,dt \approx \mathcal{V}(\sigma - \sigma_{\text{true}})$.

Girsanov (Lesson 25)

- $Z_t = e^{-\theta W_t - \frac{1}{2}\theta^2 t}$; $dZ = -\theta Z\,dW$; $E[Z_T] = 1$; $E[Z_T^2] = e^{\theta^2 T}$.
- $\tilde{P}(A) = E[Z_T \mathbf{1}_A]$; $\tilde{W}_t = W_t + \theta t$ is Brownian under $\tilde{P}$.
- $\theta = (\mu - r)/\sigma \Rightarrow dS = rS\,dt + \sigma S\,d\tilde{W}$.

Risk-neutral pricing (Lesson 26)

- $d(D_tS_t) = \sigma D_t S_t\,d\tilde{W}_t$; $d(D_tX_t) = \Delta_t\,d(D_tS_t)$.
- $V_t = \tilde{E}[e^{-r(T-t)}V_T \mid \mathcal{F}_t]$.
- Representation: $dM_t = \Gamma_t\,d\tilde{W}_t$; hedge $\Delta_t = \Gamma_t/(\sigma D_t S_t)$.

Numéraire (Lesson 27)

- $Z_T^S = e^{-rT}S_T/S_0$; $W_t^S = \tilde{W}_t - \sigma t$; $dS = (r + \sigma^2)S\,dt + \sigma S\,dW^S$.
- $V_0 = S_0 E^S[V_T/S_T]$; $C = S_0 P^S(S_T > K) - Ke^{-rT}\tilde{P}(S_T > K)$.
- Margrabe: $V_0 = S_0^1 N(d_1) - S_0^2 N(d_2)$, $\sigma_x^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, $d_{1,2} = \dfrac{\ln(S_0^1/S_0^2) \pm \frac{1}{2}\sigma_x^2 T}{\sigma_x\sqrt{T}}$.

Feynman-Kac (Lesson 28)

- $g(t,x) = E[e^{-r(T-t)}h(X_T) \mid X_t = x]$ solves $g_t + ag_x + \frac{1}{2}b^2 g_{xx} = rg$, $g(T,x) = h(x)$.
- Vasicek: $P = e^{A - Br}$, $B = (1 - e^{-\kappa\tau})/\kappa$, $A = (\bar{r} - \frac{s^2}{2\kappa^2})(B - \tau) - \frac{s^2 B^2}{4\kappa}$.

Finite differences (Lesson 29)

- Central difference error $\frac{1}{6}h^2 f'''$; second difference error $\frac{1}{12}h^2 f''''$.
- $U_\tau = \nu U_y + \frac{1}{2}\sigma^2 U_{yy} - rU$.
- Explicit: $U_i^{n+1} = (p_u U_{i+1}^n + p_m U_i^n + p_d U_{i-1}^n)/(1 + r\Delta\tau)$; $p_{u,d} = \Delta\tau(\frac{\sigma^2}{2\Delta y^2} \pm \frac{\nu}{2\Delta y})$; $p_m = 1 - c$.
- Stable when $c = \sigma^2\Delta\tau/\Delta y^2 \le 1$. Growth per step of a zigzag: explicit $|1 - 2c|$; implicit $1/(1 + 2c)$; Crank-Nicolson $|1 - c|/(1 + c)$.

Choosing a method (Lesson 30)

- Cost against error ε: tree ε^{-2}, explicit grid $\varepsilon^{-3/2}$, Monte Carlo ε^{-2} at any dimension. A d-asset grid: exponential in d.

- Delta: pathwise $e^{-rT}\mathbf{1}\{S_T > K\}S_T/S_0$; likelihood ratio: payoff times $Z/(S_0\sigma\sqrt{T})$.

The course's numbers

Share price today, S_0	100
Strike, K	105
Risk-free rate, r (continuous)	3%
Volatility, σ	20%
Expected return in Part 4, μ	8%
Market price of risk, θ	0.25
Horizon, T	1 year
d_+, d_-	0.00605, −0.19395
$N(d_+)$, $N(d_-)$	0.50241, 0.42311
Call, put	7.1281, 9.0248
American put	about 9.478

Glossary

Antithetic variates. Pairing each random draw Z with $-Z$, to reduce Monte Carlo's variance.

Black-Scholes equation. $V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV$.

Complete market. One in which every payoff can be replicated.

Control variate. A related quantity with a known mean, used to cancel part of Monte Carlo's noise.

Crank-Nicolson. A finite-difference scheme averaging the explicit and implicit schemes.

Curse of dimensionality. A grid's size grows exponentially with the number of dimensions.

Delta hedging. Holding V_S shares against an option, rebalanced over time.

Density process. Z_t: the running weight that defines a change of probability.

Digital option. Pays a fixed amount (cash digital) or one share (asset digital) if the option finishes in the money.

Equivalent probabilities. Probabilities that agree on which events are impossible.

Exchange option. The right to swap one asset for another; priced by Margrabe's formula.

Explicit scheme. Each new grid value is computed directly from old ones. Stable only for $c \leq 1$.

Feynman-Kac theorem. The link between expectations of a diffusion and solutions of a PDE.

Fundamental theorems of asset pricing. A risk-neutral probability means no arbitrage. If it's unique, the market is complete.

Gamma. The second derivative of the option's value with respect to the share.

Girsanov's theorem. Under $\tilde{P}$ with weight Z_T, $W_t + \theta t$ is a Brownian motion.

Implicit scheme. Differences use the new time; each step needs a tridiagonal solve. No stability limit.

Implied volatility. The σ that makes the formula match a market price.

Likelihood ratio method. Computing a Greek by differentiating the density rather than the payoff.

Longstaff-Schwartz. Monte Carlo for early exercise, estimating continuation values by regression.

Margrabe's formula. The price of an exchange option.

Market price of risk. $\theta = (\mu - r)/\sigma$; the Sharpe ratio.

Markov process. One whose future depends only on the present value.

Martingale representation theorem. On a world driven by one Brownian motion, every martingale is an Itô integral against it.

Numéraire. The unit in which wealth is counted.

Pathwise method. Computing a Greek by differentiating the payoff along each path.

Radon-Nikodym derivative. The weight Z_T relating two equivalent probabilities.

Richardson extrapolation. Combining two sizes to cancel a known error rate.

Risk-neutral probability. A probability under which discounted asset prices are martingales.

Self-financing portfolio. One with no money added or removed after the start.

Share measure. The probability P^S that goes with the share as numéraire.

Sharpe ratio. Excess return per unit of volatility.

Standard error. The standard deviation of a Monte Carlo estimate, $s/\sqrt{M}$.

Stencil. The set of neighbouring grid points that a scheme uses.

Theta. The rate of change of the option's value with calendar time.

Thomas algorithm. The fast way to solve a tridiagonal system.

Trinomial tree. A lattice with three branches per step; the explicit scheme by another name.

Vasicek model. $dr = \kappa(\bar{r} - r)\,dt + s\,dW$.

Vega. The derivative of the option's value with respect to volatility.

Further reading

- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*, Chapters 4 to 6 and 9: Black-Scholes, risk-neutral pricing, Feynman-Kac and numéraires with full rigour.
- John C. Hull, *Options, Futures, and Other Derivatives*: the practitioner's standard, with trees and Greeks.
- Paul Glasserman, *Monte Carlo Methods in Financial Engineering*: variance reduction, Greeks, and American options on paths.
- Daniel J. Duffy, *Finite Difference Methods in Financial Engineering*: schemes and stability in depth.
- Fischer Black and Myron Scholes, "The Pricing of Options and Corporate Liabilities" (1973), and Robert C. Merton, "Theory of Rational Option Pricing" (1973): the original papers.
- Francis A. Longstaff and Eduardo S. Schwartz, "Valuing American Options by Simulation: A Simple Least-Squares Approach" (2001).
- William Margrabe, "The Value of an Option to Exchange One Asset for Another" (1978).